

FIRM SIZE AND WAGES

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Contents

Abstract	2166
JEL codes	2166
1. A diversity of jobs	2167
2. The firm in product and labor markets	2169
2.1. The size distribution of firms and establishments	2169
2.2. Wages in relation to firm and establishment size	2172
2.3. Worker characteristics and the skill mix of the workforce	2179
2.4. Self-selection and the size-wage gap	2182
3. Some behavioral explanations	2184
3.1. Monitoring costs and entrepreneurial ability	2184
3.2. Shirking and efficiency wages	2186
3.3. Wages and the ability to pay	2187
4. The productivity hypothesis	2189
4.1. Effort, productivity, and the disutility of work	2190
4.2. Economies of massed reserves	2192
4.3. Capital, technology, and labor productivity	2195
4.4. The organization of production	2201
4.5. Durable jobs and human capital	2204
5. Why does employer size matter?	2207
Appendix A. On the relationship of wages to firm age	2209
References	2211

Abstract

Jobs differ along many dimensions including firm size. The wage gap due to firm size of 35% is comparable to the gender wage gap of 36% for men over women and greater than the wage gap of 14% for whites over black employees. The size-wage premium is larger for men and varies across industries. It is larger in the US than in other industrialized countries. Large firms demand a higher quality of labor defined by such observable characteristics as education, job tenure, and a higher fraction of full-time workers. Part 3 examines three behavioral explanations. (1) Productive employees are matched with able entrepreneurs to minimize the sum of wages and monitoring costs. (2) Big firms pay efficiency wages to deter shirking. (3) Big firms adopt a discretionary wage policy to share rents, or in Slichter's words, "Wages over a considerable range reflect managerial discretion. When management can easily afford to pay high wages, they tend to do so." We advance a productivity hypothesis. A large organization sets a higher performance standard that raises labor productivity but has to be supported by a compensating wage difference. In the service industries, the pace of work depends on the customer arrival rate. The economies of massed reserves generates a positive wage-size profile. The capital/labor ratio is higher in bigger firms which also are early in adopting new technologies. Both forces raise the demand for skilled labor where skill and productivity are often unobservable traits. Production organized around teams calls for conformance to common work rules which result in paying rents to infra-marginal team members. The odds of survival are higher for big firms which enable them to "produce" more durable employees who are more productive because they get more training. Firm size is a function of external market forces, technology, managerial decisions, and luck. The surplus of revenues over labor costs per employee is positively related to firm size for three reasons, lower prices for non-labor inputs, possibly greater market power, or larger overhead costs to amortize the sunk costs for capital and firm-specific work force. Rent sharing cannot be dismissed as an explanation for the wage-size premium. Taxation and regulation can also affect the size distribution of firms. The organization of work and the selection of employees (whose productive traits are not always observable) are responsible for the positive relation between wages and employer size.

JEL codes: J3

We started out to find the relation of the concentration of industry to the changing status of the laborer, and we proposed to investigate that relation from the four points of view of (1) the rate of wages, (2) the amount of employment, (3) the continuity of employment, and (4) the length of the working day. Our investigation has yielded the definite result that, as the size of the establishment increases, the condition of the laborer improves in all directions – his wages rise, he is employed a greater number of days in a year, his employment varies less from month to month, and his hours of labor, per day, decrease. (Moore, 1911, p. 153)

A labor market in which wages depend on employer size means that jobs are different. A job is described by a vector of variables including the rate of pay, the length of the workweek, the stability of employment, health and injury risks, the nature of the tasks,

opportunities for promotion, and the characteristics of the workplace. Jobs can be classified by occupation, industry, ownership (public versus private), geographic location, or employer size. This chapter is concerned with the size dimension.

Large firms with more than 500 employees are few in number, comprising less than one-third of 1% of all firms but providing jobs for nearly half of all employed persons. Because firm sizes are related to industrial affiliation, the sectoral shifts over the last half century have affected the size distribution. Differences in the output mix, production technologies, and government regulations account for much of the differences in the size distribution of firms across countries. The wage differential between small and large establishments which Moore discovered is substantial and pervasive. The wage gap due to firm size is approximately equal to the gender wage gap and larger than that associated with unionism or race. The magnitude of the size–wage gap is investigated in Section 2. The literature surveyed by Brown and Medoff (1989), Groshen (1991), and Troske (1994) offers several explanations for a positive relation between wages and employer size which we call the wage–size profile. The explanations that appeal to monitoring costs, efficiency wages to deter shirking, and rent-sharing are critically evaluated in Section 3.

In Section 4, the chapter advances the elements of a productivity hypothesis. The mandated effort levels or performance standards are usually unobservable aspects of the job package. Greater effort can raise labor productivity, but to elicit it, an employer will have to pay a higher wage. There are increasing returns to establishment and firm size due to technical economies as well as agglomeration effects. The exploitation of these economies is facilitated by organizing work around teams. Wage premiums have to be paid to attract workers who will comply with the administrative rules of large organizations. The volume production of standardized goods and services is evidently more economically achieved by demanding more durable capital goods as well as durable employees who are “produced” by investments in human capital. Training raises labor productivity, and sharing the returns to training generates the upward sloping wage–size profile. Ehrenberg and Smith (1997, p. 409) identify the relationship of wages to employer size as one of three puzzles looking for a solution. The hypotheses and evidence reviewed in this chapter will bring us a little closer to a satisfactory explanation.

1. A diversity of jobs

A wage rate is not the price of labor. Adam Smith observed that competition in the labor market does not tend to an equality of wages but rather to an equality of the net advantages of employment. Marshall listed four peculiarities that distinguish labor from other factors of production: (1) labor services were perishable; (2) conditions at the workplace mattered; (3) labor was at a disadvantage in bargaining;¹ (4) it takes time to acquire skills.

¹ Marshall (1920, p. 559) wrote, “It is, however, certain that manual labor as a class are at a disadvantage in bargaining... and the disadvantage is likely to be cumulative in its effect.” It not only lowered his current wage, but it also reduced his efficiency in future periods.

The labor market is not a single exchange. It is a multiplicity of “markets” in which a job is defined by a worker-firm attachment. Jobs differ not only in the tasks that an employee must perform, but also in the obligations of an employer to her employees. A new entrant often takes a starter job that introduces him to the world of work, teaches him to be punctual, presentable, and responsive to instructions. It also provides him with a track record. A second type can be called transitional filler jobs that yield a cash flow in return for labor time. The people who take these jobs rarely intend to stay for long. Their employers know this and design these jobs anticipating high turnover rates. A majority of all jobs are permanent employment relations supported by implicit or explicit contracts. They correspond to what Hall (1982) called “lifetime jobs” that last for 15 or more years. Some are located in small firms, but most of the lifetime jobs are in units where work is organized around teams.

Farber (1997) identified four dimensions that differentiated jobs – task, employer, location, and individual. This grid defines too many jobs for analytic or empirical research. For his empirical analysis, Farber turned to three dimensions: (a) the wage rate; (b) full- or part-time work; and (c) presence or absence of employer provided health insurance. Farber cited a report prepared by the Council of Economic Advisors dated April 23, 1996 in which jobs were classified by 45 occupations and 22 major industries. The mean wage for each occupation/industry cell was assigned to all employees in that O/I cell. A “good” job was defined as a high-wage job paying a wage above the nation-wide average. Using this criterion, a majority of the new jobs created between February 1994 and February 1996 were “good” jobs located in a high wage labor market defined by industry and occupation.

What determines the wage of a stacker working in a sawmill? The wage rate is not the price paid by the employer, nor is it the compensation received by the employee. It is, however, a useful analytic concept to describe the equilibrium in a labor market. The workers in the same “market” are presumably close substitutes for one another. Occupation hopefully controls for training and experience. Is industrial affiliation the proxy for working conditions? In practice, the boundaries of a “market” are fuzzy. Stigler and Sherwin (1985) examined the co-movements of prices over time to define a “market”. Gasoline Service Stations in Denver and Omaha are placed in the same market if prices in the two cities move together. Labor economists usually accept geographic proximity to define a local labor market. They tacitly assume that the wages of lawyers and of nurses are determined in different markets. High and low wage labor markets are evidently differentiated by the skills of the workers. Doeringer and Piore (1971) distinguished between internal and secondary labor markets based on the characteristics of the employer. The size of the employer measured by assets, sales, or employment can serve as a variable differentiating related labor markets. Jobs in large firms pay higher wages and would thus be classified as “good jobs”. What is a good job for one person might not be for another. The rank ordering of jobs has to account for varying preferences for other dimensions of a job package such as the disutility of work, the effort bargain, or compliance with strict orders. Although size is not a sufficient statistic to identify distinct “labor markets”, the data reveal significant differences in the nature of employment relations related to

employer size. The object of this chapter is to examine these differences to explain the positive relation between firm size and wages.

2. The firm in product and labor markets

A firm assembles resources and organizes production. But what is a firm? In a high information and low transaction cost world, everyone ought to have access to the same technologies and equal opportunities to bid for inputs. Firms in a given industry ought to be alike and achieve the same optimum size and structure. This implication is clearly refuted by the data which is briefly reviewed in Section 2.1. Further, the wages and working conditions of employees are systematically related to employer size. The matching of more able workers with larger employers is examined in Sections 2.3 and 2.4. Worker characteristics and sorting, however, can explain only a part of the relation.

2.1. The size distribution of firms and establishments

At the outset, two issues have to be addressed. First, what is the appropriate unit of analysis, the firm or establishment? Theory is of little help, and the data indicate that both matter. Second, how do we measure size, by assets, sales (a proxy for output), or employment? The capital/labor, K/L , ratio varies widely across industries because of differences in the technology of production. Within each industry, the K/L ratio is higher for larger firms. As a consequence, the four or eight firm concentration ratio based on assets is always larger than that based on sales or employment. Comparisons across industries or over time are complicated when one uses a sales measure of size. Variations in labor productivity over time as well as across firms and industries make employment an imperfect measure of size, but given its ready availability, we shall adopt this measure.

In 1992, there were over 5 million firms in the private sector of the US economy. Most firms are very small; 644 thousand had no employees, and 88.2% of those with workers had fewer than 20 employees. The large firms attain their size by controlling several establishments or plants. The big manufacturing companies owned an average of 10.1 establishments, while the large retail trade firms controlled an average of 111.1 establishments. Across all industries, large firms provided nearly half, 47.0%, of the 92.8 million private sector jobs. The size distributions of firms F , establishments or plants E , and employment N in 1992 and 1982 are shown in Table 1 for all industries, Manufacturing, and Retail Trade.² The share of industry-wide employment residing in big firms varied from lows of 11.8% in Agriculture and 11.4% in Construction to highs of 64.5% in Transportation/Communications/Utilities and 61.8% in Manufacturing; see Table 2. The growth of the economy in the postwar years was accompanied by a rightward shift in the

² These data come from the Enterprise Statistics assembled by the Census Bureau. They may differ from the establishment surveys conducted by the Bureau of Labor Statistics. We thank Elaine Manual for providing us with these statistics.

size distributions of firms and plants, at least through 1970. The shift in the industrial structure of the economy away from goods towards services pushes the size distribution toward the left. The share of total employment located in the Distributive Trades and Personal Services rose from 33.3% in 1957 to 49.8% in 1992.

Employment, by sector			
	1957	1979	1992
Employment (000)	52853	89823	108437
1. Manufacturing (%)	32.49	23.42	16.77
2. Wholesale Trade (%)	5.75	5.81	5.52
3. Retail Trade (%)	14.85	16.67	17.65
4. Personal Services (%)	12.69	19.05	26.65
5. Government (%)	14.41	17.75	17.13

In addition to these sectoral shifts, the size distribution is affected by market regulations and technology. Loveman and Sengenberger (1991) reported that bread is mass produced in Norway and Sweden, but there are over 30,000 bread bakeries in Germany. The latter is attributed in part to a World War I regulation that prohibited baking bread at night. The percentage of total employment in large firms was 46% in Germany and 17% in Italy. Davis and Henrekson (1997) estimated that Sweden led the list of OECD countries with 60.4% of all employees in firms with 500 or more workers. Part of the concentration in Sweden is due to the dominance of the public sector. Health care and social services are provided by public sector employees whose pay is determined by some process other than a competitive labor market. Loveman and Sengenberger (1991) observed that there is a re-emergence of small and medium sized firms. The time series data for the OECD countries exhibit a V-shaped pattern for the share of employment in small enterprises with fewer than a hundred employees; the trough occurs around 1969–1970. The reasons for the recent growth in the small firm share include among others, privatization and technological advances which raised the efficiency of small units. Data for the US are shown below for All Industries, Manufacturing, and Retail Trade.³

Percentage of total employment in small firms			
Year	All Industries	Manufacturing	Retail Trade
1977	40.13	16.17	53.56
1982	45.51	17.56	51.98
1987	42.46	18.44	45.53
1992	38.66	22.08	43.53

³ The US data come from the Enterprise Statistics where a small firm is one with less than 100 employees. Data for the OECD countries are reported in Table 3 of Loveman and Sengenberger (1991, pp. 7–8), while the size distribution for manufacturing establishments is reported in their Table 4.

Table 1
Percentage distribution by size and industry, 1992 and 1982^a

Industry and year	Number	Percent in firms with an employment of:			
		1-19	20-99	100-499	500+
<i>A. 1992 All Industries</i>					
Firms F	4451	88.26	9.87	1.55	0.32
Plants E	5673	70.64	11.19	5.00	13.17
Employees N	92826	20.22	18.44	14.34	47.00
Annual Pay W	24249	20156	21551	22407	27632
<i>M. 1992 Manufacturing</i>					
Firms F	300	71.48	21.58	5.42	1.53
Plants E	359	51.79	19.38	7.90	12.92
Employees N	18167	7.44	14.64	16.16	61.76
Annual Pay W	21916	24992	25998	34680	30909
<i>RT. 1992 Retail Trade</i>					
Firms F	996	86.83	11.58	1.30	0.30
Plants E	1442	59.90	11.95	5.81	22.34
Employees N	19681	22.53	21.00	10.48	45.99
Annual Pay W	11964	17900	24549	23821	18817
<i>A. 1982 All Industries</i>					
Firms F	3997	90.74	8.01	1.06	0.18
Plants E	55355	79.96	8.25	3.01	8.72
Employees N	61462	25.69	19.82	13.08	41.41
Annual Pay W	10837	12001	13749	18522	14631
<i>M. 1982 Manufacturing</i>					
Firms F	278	71.35	22.35	5.06	1.24
Plants E	483	41.80	14.70	7.27	36.24
Employees N	22008	5.70	11.86	12.74	69.69
Annual Pay W	14370	15599	16277	22032	20099
<i>RT. 1982 Retail Trade</i>					
Firms F	947	89.08	9.74	1.04	0.13
Plants E	1289	68.33	11.54	4.82	15.31
Employees N	14845	28.87	23.11	9.68	38.34
Annual Pay W	7849	8767	8886	10400	9140

^a Source: Enterprise Statistics, Census Bureau.

The upward trend in the small firm share is only observed in Manufacturing where the computer and other technical advances apparently have reduced the optimum firm size where size is measured by employment. Firm size is surely endogenous. Size depends on

Table 2
Employment and annual payroll by major industry, 1992^a

	In firms with an employment of:			
	Total	1-19	500+	Ratio ^b
<i>A. Employment (000)</i>				
0. Total Private	92826	18773	43625	47.0
1. Agriculture	594	332	70	11.8
2. Mining	650	82	393	60.4
3. Construction	4502	2041	515	11.4
4. Manufacturing	18167	1351	11220	61.8
5. Trans/Communication	5521	662	3564	64.5
6. Wholesale Trade	6095	1548	2030	33.3
7. Retail Trade	19681	4435	9052	46.0
8. FIRE	6904	1203	3941	57.1
9. Services	30666	7082	12840	41.9
<i>B. Annual payroll per employee (dollars)</i>				
0. Total Private	24249	21297	27632	1.297
1. Agriculture	16285	16787	19390	1.155
2. Mining	39179	25901	45545	1.758
3. Construction	26567	23215	35202	1.516
4. Manufacturing	30909	23187	34680	1.496
5. Trans/Communication	31648	20443	36592	1.790
6. Wholesale Trade	31085	27449	37553	1.368
7. Retail Trade	12995	12218	13442	1.100
8. FIRE	31742	25359	35460	1.398
9. Services	22670	24208	24213	1.000

^a Source: Enterprise Statistics, US Bureau of the Census.

^b Ratio for (A), employment, represents percentage of industry employment located in firms with 500 or more employees. Ratio for (B), annual payroll, is the ratio of pay in big firms over pay in small firms.

the market prices for inputs and the technologies chosen by the firm. This is not the place to discuss the reasons why some firms shrink and die, while others reach gigantic size. One thing is clear, the large firms in an industry evidently occupy a different place in product and input markets than their smaller competitors.

2.2. Wages in relation to firm and establishment size

Wages are higher in larger firms. This empirical regularity was discovered by Moore (1911) who conducted a statistical study. He wanted to obtain wage data that controlled for the sex and age of the worker as well as the location of the worksite – geographic region, city versus country, and industry; see Moore (1911, p. 140). He settled for data on the daily wages of Italian working women in textile mills classified by age and establishment size. His data on daily wages are shown in Table 3. Wages at the largest plants with

Table 3
Mean daily wages and mean ages of Italian women^a

Age group	In establishments with an employment of:			
	<20	20–99	100–499	500+
<i>A. Daily wages in lire</i>				
15–20	0.87	0.93	1.04	1.24
20–35	1.09	1.10	1.21	1.50
35–55	1.05	1.12	1.17	1.48
>55	0.92	0.98	0.98	1.16
Mean wage	1.002	1.030	1.130	1.385
<i>B. Mean ages</i>				
15–20	17.32	17.30	17.30	17.31
20–35	25.83	25.33	25.22	25.34
35–55	44.43	44.72	44.89	43.63
>55	58.20	57.53	57.58	57.49
All ages	28.23	25.63	25.14	24.32
Employment <i>N</i>	2166	32523	93566	23058
Establishments <i>E</i>	239	804	619	37
Average size <i>N/E</i>	9.06	40.45	151.16	623.19
<i>C. Regression coefficients</i>				
Original		0.039	0.141	0.392
Log		0.39	0.134	0.332

^a Source: Tables 1, 2, 4, and 5 in Moore (1911).

500 or more employees were some 38.5% above what employees received at the smallest plants. The mean wages shown in the fifth line describe a concave relationship; the arc elasticities were 0.022, 0.081, and 0.167 across the four size categories. It is also evident that wages are related to mean age. We estimated a wage equation including indicator variables for plant size and a quadratic in experience defined as $X = (\text{Age} - 16)$.⁴ The addition of X and X^2 not only improves the goodness of fit (the R^2 climbs from 0.59 to 0.97), but also increases the slope of the wage–size profile. Given work experience, workers in the largest plants earned 39.2% more than their peers in small textile mills. Are workers at large plants on a higher plane of well-being than others employed at small plants? His statistical investigations led Moore to the following conclusion (p. 163), “...as

⁴ The wage equation was, $W_{ij} = SB + c_1X_{ij} + c_2X_{ij}^2 + e_{ij}$, where W_{ij} is the mean daily wage in the i th age group and the j th plant size category, S is a vector of size dummy variables, and X_{ij} is the mean years of experience of workers shown in (B). The parameters were estimated by weighted least squares using the employment weights reported in Appendix Table 3 of Moore (1911, p. 166). The coefficients of the size dummies are shown in Table 3. The quadratic in experience was for the original values, $W = 0.0288X - 0.000738X^2$. For the log equation, we had $\ln W = 0.0254X - 0.000650X^2$.

the size of the establishment increases, the condition of the laborer improves in all directions – his wages rise, he is employed a greater number of days in a year, his employment varies less from month to month, and his hours of labor, per day, decrease.”

The relations discovered by Moore were not unique to Italian textile mills. They prevail in nearly every labor market. Lester (1967) reported that average hourly earnings in large establishments were 20–25% above the average hourly earnings in small establishments belonging to the same industry. Mellow (1982) analyzed the May 1979 CPS data. Hourly earnings were positively related to both firm and establishment size even after controlling for worker characteristics and industrial affiliation. Brown et al. (1990, p. 30) reported patterns from the May 1983 CPS. Hourly wages in firms with 500 or more workers were 35% above wages in firms with less than 25 employees. The size–wage premium of 35% was of similar magnitude to the 36% wage gap between men and women, and exceeded the wage gap of 29% for union over non-union workers and 14% for white over black employees.

The bivariate associations between firm size and selected variables taken from the May 1993 CPS are shown in Table 4. Hourly wages of men, defined as usual weekly earnings

Table 4
Wages and related variables by firm size and sex, 1993^a

Variable	F1 1–24	F2 25–99	F3 100–499	F4 500–999	F5 1000+	Ratio ^b
Females						
Sample size	2120	1087	1081	442	3167	
Wage	8.203	9.052	10.114	10.525	10.683	1.302
Tenure	5.664	6.093	6.843	7.212	8.128	1.435
Education	12.698	12.807	13.109	13.239	13.137	1.035
White	91.698	88.960	88.714	87.330	85.475	0.932
Married	58.726	56.486	56.152	56.335	54.500	0.928
Part-time	39.906	24.103	21.462	19.231	23.745	595
Union ^c	1.063	4.019	7.034	11.848	13.583	12.778
Pension ^d	14.554	28.044	48.293	50.856	61.544	4.229
Males						
Sample size	2144	1302	1189	451	3698	
Wage	10.289	12.381	13.459	13.528	14.951	1.452
Tenure	6.338	7.030	8.089	9.125	11.246	1.774
Education	12.515	12.786	13.193	13.181	13.494	1.078
White	90.951	90.860	91.926	89.135	88.886	0.977
Married	55.364	61.290	63.751	66.962	66.820	1.207
Part-time	18.470	8.372	7.653	7.539	9.708	0.536
Union	5.005	10.925	13.832	18.307	24.784	4.952
Pension	12.748	38.591	56.495	61.575	73.604	5.774

^a Source: April 1993 Current Population Survey

^b Ratio = F5/F1.

^c Union = 1 if either a union member or covered by a union contract.

^d Pension = 1 if covered by a pension or retirement plan.

Table 5
Hourly wages by sex and firm size^a

Year and firm size	Male	Female	Both	Ratio ^b
<i>A. 1983 Wages</i>				
F1 = 1–24	6.97	5.26	6.178	0.755
F2 = 25–99	8.54	5.95	7.384	0.697
F3 = 100–499	9.57	6.51	8.132	0.680
F4 = 500–999	9.79	6.76	8.272	0.691
F5 = 1000+	11.34	7.28	9.643	0.642
<i>B. 1979 Wages</i>				
F1 = 1–24	5.646	4.052	4.880	0.718
F2 = 25–99	6.689	4.239	5.643	0.634
F3 = 100–499	7.427	4.689	6.208	0.631
F4 = 500–999	7.82	4.714	6.363	0.603
F5 = 1000+	8.452	5.235	7.293	0.619
<i>C. Percent of employment in large firms</i>				
1993	47.2	45.7	46.5	
1983	41.2	38.1	39.8	
1979	45.9	37.4	42.3	
<i>D. Wages in small firms</i>				
1993	11.690	8.897	10.348	0.761
1983	7.901	5.693	6.884	0.721
1979	6.350	4.289	5.405	0.675
<i>E. Wages in large firms</i>				
1993	14.795	10.662	12.872	0.721
1983	11.166	7.202	9.466	0.645
1979	8.381	5.158	7.183	0.615
<i>F. Wage ratio, large/small</i>				
1993	1.266	1.198	1.244	
1983	1.413	1.265	1.375	
1979	1.321	1.203	1.329	

^a Source: Tabulated from CPS tapes.

^b Gender wage ratio of female/male wages.

divided by usual weekly hours, rise from \$10.29 in small firms with 1–24 employees, size category F1, to \$14.95 in size category F5, 1000 or more employees. The wage ratio was 1.453 compared to a ratio for female workers of 1.302. Fringes increase even faster – pension coverage climbs from 12.75 for men in small firms to 73.60% in F5. Hourly compensation including fringes is more strongly related to size than wages alone, see Brown and Medoff (1989, p. 1036). The mean duration of job tenure of those on the payroll is longer in larger firms which is consistent with a model in which larger employers

Table 6
Average hourly earnings by industry and sex (by firm size, May 1983 CPS)^a

Industry and sex	In firms with an employment of			Ratio ^b
	Total	1-24	1000+	
<i>A. Males, 1983</i>				
1. Agriculture	4.677	4.388	6.436	1.467
2. Mining	12.369	8.316	13.487	1.622
3. Construction	9.380	7.995	13.679	1.711
4. Manufacturing	10.300	7.344	11.705	1.594
5. Trans/Communication	11.541	7.761	13.096	1.687
6. Trade	7.433	6.253	8.438	1.349
7. Finance	11.696	8.437	12.588	1.492
8. Services	8.677	7.526	10.020	1.331
<i>B. Females, 1983</i>				
1. Agriculture	4.696	4.556	5.013	1.100
2. Mining	9.606	9.917	9.706	.979
3. Construction	6.687	6.344	8.262	1.302
4. Manufacturing	6.880	6.032	7.714	1.279
5. Trans/Communication	8.697	5.722	9.787	1.710
6. Trade	4.858	4.403	5.269	1.197
7. Finance	6.902	6.193	7.538	1.217
8. Services	6.656	5.955	7.759	1.303

^a Source: Tabulated from the May 1983 Current Population Survey.

^b Ratio = wages in firm size 1000 + /firm size 1-24.

provide more specific training. Employees in larger firms are slightly older and have more years of schooling. The fraction on part-time schedules is inversely related to size falling from 39.9 to 23.7% for females.

Table 5(A,B) describes the wage-size profiles based on the May CPS data for 1983 and 1979. The shape of the wage-size profile obviously differs by gender and has shifted over time. The wage ratio for males in the largest and smallest size categories {F5 over F1} was 1.627 in 1983 and 1.497 in 1979. The corresponding figures for females were 1.384 and 1.292. The wage-size premium is clearly larger for males. To the extent that small employers employ less skilled workers and the dispersion of wages expands in a downturn, it is not surprising to find a larger size-wage gap in 1983, a recession year. The last column presents the gender wage ratio, $G = (W_f/W_m)$ for each size group. Notice that the gender wage gap G declines with firm size; the relative wages of females are higher in small firms.

In the last four panels of Table 5, workers are divided into two size groups, <500 versus 500+ employees. In small firms with <500 workers, the gender wage ratio climbed from 0.675 in 1979 to 0.761 in 1993; the corresponding rise in large firms, panel E, was 0.615 to 0.721. Panel F presents the size-wage ratio, $S = (W_{F5}/W_{F1})$. For both sexes, this ratio fell from 1.329 in 1979 to 1.244 in 1993.

Table 6 reveals significant industry differences. The wage–size premium in Manufacturing, described by a wage ratio was, for male workers, $(w_L/w_S) = 11.705/7.344 = 1.594$. Notice that the size–wage premiums for female employees are smaller even within the same industry. A comparison of Tables 2 and 6 which come from different surveys reveals that there is a strong correspondence in the size–wage premiums across industries. Being employed by a large firm in Transportation/Communications/Utilities or in Manufacturing is associated with a larger size–wage premium than the size premium in Retail Trade or Services. Finance, Real Estate, and Insurance falls in between leaning closer to Manufacturing.

The relation of wages to firm size is ubiquitous. Idson and Ishii (1992) assembled data for Japan and the United States for 1988 (see Table 7). The wage ratio (w_L/w_S) describing the relative premium in large over small employers is seen to be considerably larger in Japan, in the neighborhood of 1.68 compared to around 1.30 in the United States. Loveman and Sengenberger (1991, pp. 18–19) assembled wage data for several advanced countries classified by employer size. We reproduce their wage indexes for small and

Table 7

Wages, tenure, and education by sex, 1988 (by firm size for Japan and the US)^a

	In firms with employment of:			
	10–99	100–999	1000+	Ratio ^b
<i>Japan, women</i>				
Wages ^c	925.80	1105.04	1553.98	1.679
Tenure	6.31	6.367	7.183	1.137
Education	11.709	12.015	12.421	1.061
<i>US women</i>				
Wages	7.384	8.604	9.322	1.263
Tenure	3.971	5.367	6.935	1.746
Education	12.970	13.093	13.145	1.014
<i>Japan, men</i>				
Wages	1493.25	1866.54	2520.73	1.688
Tenure	8.958	11.460	15.321	1.710
Education	11.830	12.746	12.898	1.090
<i>US men</i>				
Wages	9.929	11.638	13.038	1.313
Tenure	4.890	6.587	9.711	1.986
Education	12.781	13.255	13.460	1.053

^a Source: Idson and Ishii (1992, p. 533).

^b Ratio = wages in firm size 1000+/firm size 10–99.

^c Wages are average hourly earnings in 1988; in yen for Japan and dollars for the US.

medium sized firms as a percentage of the wages in large firms with 500 or more employees.

Wage indexes by country and firm size

Country	Year	Small 10–99	Medium 100–499
France	1978	83	86
Germany	1978	90	92
Italy	1978	85	93
Japan	1982	77	83
United States	1983	53	74

The flatter profiles in Germany and France can partially be explained by stronger trade unions and training. The concave pattern observed in the United States and Japan are not replicated in the European economies. Studies of the labor markets in Peru (Schaffner, 1996), Zimbabwe (Velenchik, 1996), and Guatemala (Funkhauser, 1997) reveal an even stronger concavity than that exhibited by the US data.

Are wages related to both firm and establishment sizes? Hourly wages for male employees classified by firm and plant size are shown in Table 8. Holding plant size constant at P1, wages rose from \$7.119 to \$9.849 from the smallest F1 to largest F5 firms, a 38.3 F-size wage premium. In the largest F5 firms, the wage gain from small to big plants was 28.3%. Separate tabulations were calculated for employees in Manufacturing and in Trade. The wage ratio across firm sizes, (W_{F5}/W_{F1}) was 1.588 in Manufacturing and 1.221 in Trade. Within the Trade sector, small plants in big firms, (P1-F5 cell) are located in Eating & Drinking Places, while big plants in big firms (P5-F5 cell) are located in Department Stores and Wholesale Trade. Brown and Medoff (1989) found that the partial effect of establishment size on wages was stronger than that of firm size.⁵ The factors responsible for the wage-size profiles are clearly not the same across industries and gender.

The wage-size relation extends across the skill spectrum. A janitor and a lathe operator earn higher wages if they are working for a large rather than a small sawmill. Doms et al. (1997) report that across all skill levels, the elasticity of the hourly wage with respect to establishment size was around +0.06. A tenfold increase in firm size is accompanied by a 16% higher hourly wage. This elasticity appears to be smaller for workers at higher skill levels. However, at the top of the skill hierarchy, we find a reversal of this pattern. The salary of the Chief Executive Officer Y is related to size measured by annual sales S via an exponential relation of the form, $Y = AS^\eta$. An elasticity of $\eta = +0.25$ means that a

⁵ Consider the regression, $W = b_0 + b_1F + b_2P + \text{error}$, where all variables, wages W , firm size F , and plant size P are in logs. Data from the May 1983 CPS showed that for females, the plant size effect is stronger than the firm size effect. Brown and Medoff combined the sexes and included a female dummy variable. The relative strengths of the two size measures fluctuates by gender and by full- versus part-time status.

Table 8

Average hourly earnings by firm and plant size (male employees by industry, May 1983 CPS)

Industry/plant size ^a	F1	F2	F3	F4	F5
<i>All industries</i>					
P1	7.119	8.686	9.297	9.617	9.849
P2		8.631	10.162	9.326	10.444
P3			9.616	10.161	11.085
P4				9.424	12.637
P5					12.637
<i>Manufacturing</i>					
P1	7.344	8.034	7.63	10.412	11.664
P2		8.425	9.44	10.398	10.8
P3			8.898	9.572	10.528
P4				9.618	11.322
P5					12.61
<i>Trade</i>					
P1	6.253	6.529	9.638	7.129	7.634
P2		7.636	8.152	7.046	8.19
P3			8.778	9.884	8.526
P4				7.839	11.774
P5					11.396

^a The five firm and plant size categories correspond to employments of 1–24, 25–99, 100–499, 500–999, and 1000+ employees.

tenfold increase in firm size leads to a 78% increase in the salary of the CEO, considerably more than the 16% size–wage premium enjoyed by hourly employees.⁶ We shall focus on the relation of size to the pay of wage and salaried employees and leave the analysis of executive compensation to others.

2.3. Worker characteristics and the skill mix of the workforce

Workers are different, and the more able command higher wages. Are the wages at larger firms a consequence of the fact that larger employers simply demand more able employees? Henry Moore tried to isolate the partial effect of size by holding constant the gender and age of the worker. Regression models and the micro data to implement them allow us

⁶ These regression equations are regularly reported by the Conference Board. Estimates for η vary across industries and over time but fall in the range from +0.25 to +0.35; see Table 2 in Baker et al. (1988). These results were also obtained by Kostiuick (1990) for several samples including a panel dataset. In fact, the relation held for the salaries of union presidents where a tenfold increase in the size of the union was associated with a 78% increase in salary.

to introduce a larger vector of variables that hopefully serve as proxies for worker ability. A standard approach is to specify two log-linear wage equations.

$$Y = SA_1 + e_1, \quad (1)$$

$$Y = SA_2 + XB_2 + e_2, \quad (2)$$

where Y is the log of hourly earnings, S is a vector of 8 dummy variables corresponding to firm/plant sizes, and X is a vector of worker characteristics identified in Table 9.⁷

The wage premiums associated with firm and plant size were converted to percentage increments in Table 10 by taking anti-logs of the parameter estimates reported in Table 9. MF and LF stand for medium and large firms with 100–499 and 1000+ employees, while SP designates a small plant with less than 25 workers. Full-time males in a small plant of medium size firms, SP/MF, earned hourly wages that were 25.7% above the mean wage of \$6.62 paid to workers in the base group, single plant firms with less than 25 employees, SP/SF. Ignoring worker characteristics, Eq. (1), workers in large plants of big firms, LP/LF, received a wage premium of 62.6%. When education, job tenure, and other worker traits are included in the wage equation, this size premium drops to 27.8% which squares with a matching model. Because big firms assemble workforces with a richer skill mix, adjusting for worker characteristics reduces the magnitude of the wage–size differential. Reference to Table 10 reveals that the premium for firm/plant size is larger for males than for females. Moreover, the relation of hourly wages to firm/plant size is observed even when the sample is restricted to full-time employees.

The estimation of size effects via dummy variables is appropriate only if the partial effects of worker characteristics on log wages are the same across size categories. If the size and worker characteristic vectors are interacted, a F -test rejects the hypothesis that the slope parameters B are the same across firm size categories. However, when separate wage equations are estimated, full-time workers in small firms with less than 100 workers earned lower wages than their counterparts in large firms with 1000 or more employees in spite of differences in mean characteristics and slope parameters.⁸

Personick and Barsky (1982) analyzed the data from the white collar pay surveys. Workers in the same narrowly defined occupations such as accountant 3, secretary 2, or computer operator 3 are likely to be more homogeneous than workers with the same years of schooling, job tenure, and industrial affiliation. These data ought therefore to be superior to the CPS to measure the impact of firm size for otherwise equally qualified employ-

⁷ Eq. (1) is a log counterpart of Table 8, but with workers classified into fewer firm and plant size categories. The base group corresponds to small firms with less than 25 employees. The remaining eight groups are defined by the four firm size categories identified in Table 4 and two plant size categories (small plants with 1–24 employees and large plants with 25 or more employees).

⁸ Oi (1991) identified sub-samples which held constant full- versus part-time status, gender, age, and firm/plant size. The exercise was not wholly satisfactory, but the results lend additional support to the positive relation between firm/plant size and wages.

Table 9
Wage equations for full-time employees by sex, 1983^{a,b}

Variable	Male employees			Female employees		
	Mean	B	t	Mean	B	t
<i>A. Firm/plant size dummies^c</i>						
F2SP	0.030	0.110	3.96	0.032	0.088	3.06
F3SP	0.025	0.092	3.04	0.027	0.127	4.06
F4SP	0.008	0.147	2.76	0.007	0.048	0.83
F5SP	0.051	0.117	5.17	0.040	0.131	4.96
F2LP	0.115	0.087	5.32	0.116	0.075	4.41
F3LP	0.109	0.142	8.38	0.124	0.127	7.50
F4LP	0.043	0.134	5.53	0.055	0.160	7.00
F5LP	0.353	0.245	17.90	0.316	0.232	17.00
<i>B. Worker/job characteristics</i>						
Education	12.915	0.063	33.45	12.684	0.064	26.77
Ten	8.205	0.020	12.09	5.537	0.028	14.17
Ten-2	145.516	-0.040e-2	-8.01	72.606	-0.058e-2	-8.05
Exp	18.452	0.025	16.02	17.772	0.012	8.22
Exp-2	496.391	-0.043e-2	-13.10	473.881	-0.027e-2	-8.35
Married	0.744	0.122	10.52	0.629	0.003	0.30
Black	0.055	-0.170	-8.14	0.078	-0.100	-5.33
SMSA	0.374	0.122	11.48	0.390	0.134	13.16
South	0.280	-0.048	-4.64	0.292	-0.047	-4.29
<i>C. Industrial affiliation</i>						
Agriculture	0.025	-0.351	-11.28	0.005	-0.170	-2.40
Mining	0.024	0.193	6.31	0.005	0.326	4.69
Construction	0.084	0.186	9.91	0.012	0.079	1.70
TCU (Utilities)	0.094	0.103	6.08	0.055	0.161	6.86
Trade	0.216	-0.129	-9.53	0.240	-0.190	-12.44
Finance	0.055	0.031	1.43	0.119	-0.006	-0.35
Service	0.162	-0.112	-7.49	0.350	-0.026	-1.84
<i>Summary statistics</i>						
ln AHE	2.155			1.777		
R-square	0.4064			0.3352		
N	7833			5973		

^a Source: May 1983 CPS.

^b Dependent variable is ln(average hourly earnings).

^c F2-F5 correspond to firm size categories 25-99, 100-499, 500-999, 1000+; SP, LP correspond to small plants (1-24) and larger plants (25+), respectively.

ees. For most of the occupations identified in their Table 1 (p. 24), wages are fairly stable for firms with less than 10,000 employees but rise sharply thereafter. Computer operators 3 in firms with 50 thousand employees or more earned 24% more than those in the smallest

Table 10
Percentage wage differentials by firm and plant size, 1983^a

Firm/plant size ^b	Full-time employees			All employees	
	(1)	(2)	(3)	(1)	(2)
<i>Male employees: percentage wage (differential)</i>					
MF/SP	25.7	9.6	4.0	31.7	5.5
LF/SP	36.5	12.4	4.1	39.1	3.7
MF/LP	32.8	15.3	9.2	39.8	8.4
LF/LP	62.6	27.8	17.4	73.2	16.3
Mean wage	8.62	8.63	8.63	8.02	8.02
Base wage ^c	6.62				
No. in sample	7857	7833	7833	8731	8705
R-square	0.1368	0.4064	0.4268	0.1529	0.5042
<i>Female employees: percentage wage (differential)</i>					
MF/SP	19.7	13.5	9.9	24.5	11.5
LF/SP	21.7	14.0	7.6	19.7	8.5
MF/LP	22.4	13.5	9.0	28.4	0.2
LF/LP	41.2	26.1	16.8	46.8	17.6
Mean wage	5.91	5.91	5.91	5.55	5.55
Base wage	5.55			4.59	
No. in sample	5998	5973	5973	7193	7164
R-square	0.0911	0.3352	0.3558	0.0978	0.3824

^a Notes: In Eq. (1), $\ln AHE$ is regressed on 8 firm/plant size dummy variables. Eq. (2) includes education, tenure, experience and industrial affiliation. Part-time employment, union and pension are included in Eq. (3). The percentage differential is the anti-log of the regression coefficient minus one, and expressed as a percentage.

^b MF/SP denotes a medium size firm with 100–499 employees and a small plant with 1–24 employees. LF/LP stands for a large firm with 1000 or more employees and large plants with 25 or more employees.

^c The base wage is the anti-log of the intercept corresponding to small firms and small plants with 1–24 employees. It is reported only for Eq. (1).

firms; the wage increment for Draftsmen 4 was 16%. To sum up, large employers demand more productive employees where “productive” is described by the usual proxies, education and experience. However, even after controlling for worker characteristics, the data support the presence of a positively inclined wage–size profile.

2.4. Self-selection and the size-wage gap

Workers are not randomly distributed across the size spectrum. If unobservable differences in productivity affect the allocation, neglect of these could impart a bias in the estimate of the effect of employer size on wages. Suppose that unobserved productivity is person-specific and unrelated to the identity of the employer. A fixed-effects model in which we take first-differences will yield a consistent estimate of the size effects. Let $[W_t, X_t, \delta, \varepsilon_t]$

denote a vector of wages, exogenous variables including establishment size, person-specific productivity effects, and a random variable in year t ; no t subscript appears for the person-specific productivity effects which are assumed to be constant over time and across employers.

$$W_t = X_t\beta + \delta + \varepsilon_t. \quad (3)$$

Differencing eliminates the unobserved person effects.

$$\Delta W = W_{t+1} - W_t = \Delta x\beta + \Delta\varepsilon. \quad (4)$$

Dunn (1980) applied this model to a sample of 200 blue-collar workers who changed employers. Workers who moved to larger employers enjoyed wage gains. The methodology was applied to other samples by Brown and Medoff (1989) and by Evans and Leighton (1989). The size effects in these fixed effects models continued to be significant even though differencing attenuated the magnitude of the coefficient of firm size. Differencing, pushes the coefficients of size and other explanatory variables toward zero because it magnifies the effect of measurement errors. The validity of a fixed effect model was questioned by Gibbons and Katz (1992). The value of unmeasured ability need not be the same across employers. Mobility could achieve superior job matches, so that the smaller size–wage gap found using fixed effects may understate the extent to which size–wage differentials are due to unobserved attributes.

Idson and Feaster (1990) try to adjust for a selection bias by first estimating an ordered probit model to predict the size category in which a worker will be employed. This equation yields estimates of the truncated means, λ_{ij} for the i th person in category j , which is then included in the wage equation. The methodology is described by Idson and Feaster (1990). Applying this methodology to 1979 CPS data for men indicated a significant positive selection bias in small firms and negative selection bias in large firms.⁹ Unobserved traits that would raise men's wages also made it more likely that they would be located in firms in the smaller size groups. The mean wages of workers in a small size category are thus higher than the mean wages that would have prevailed if workers had been randomly allocated to size categories. At the other end of the size spectrum, the men who self-selected into big firms possessed unobserved traits that depressed wages. Adjustment for the selection bias thus magnifies the size–wage gap, i.e., a random allocation of workers between large and small firms would have increased the size–wage premium. Idson and Feaster argued that this outcome is consistent with a conjecture by Stigler (1962), namely, workers with more ambition and energy would do better in a small firm where their performance will be noticed and rewarded. The procedure used to adjust the size–wage gap for self-selection into different size categories tacitly assumes that the “value” of the unobserved drive and motivation is the same across firm sizes. Attributes such as individual initiative that are productive in small firms may actually be a hindrance

⁹ Main and Reilly (1993) applied this methodology to data for British workers and found that the selection bias was negligible.

in large firms that organize production around structured teams. A random reallocation of workers across different size firms may thereby produce very different results than those predicted from this model.

The Idson–Feaster findings are consistent with the earlier work by Garen (1985). Given the difficulty of metering performance, large firms should hire workers with easily observed productivity attributes such as education. The reward to less readily observable traits such as IQ ought, therefore, to be greater in a small firm. Garen did indeed find that the return to education was higher in big firms, while small firms paid more for higher IQ scores.

Evans and Leighton (1989) argued that low quality, unstable workers are employed by small firms, a sorting pattern at variance with the Stigler conjecture. Using firm-level data, Mayo and Murray (1991) developed estimates of firm failure and employment risk. The inclusion of these risks in a wage equation eliminated the firm size effect. Winter-Ebmer (1995) used individual-level Austrian data to compute layoff probabilities. Controlling for layoff risks reduced the wage–size premium by a third, but size still exerted a significant effect on wages. These studies introduce the effect of job insecurity on wages, a variable that was not considered by Idson and Feaster. The compensating difference of higher risks for firm failure and layoff risks cannot, though, be inferred in advance. Senior and Smith voiced opposing views: “We believe after all, that nothing is so much disliked as steady, regular labor and that the opportunities of idleness afforded by an occupation of irregular employment are so much more than an equivalent for its anxiety as to reduce (such wages)...below the common average.” (Senior, 1858, p. 208)). “What he earns while he is employed, must not only maintain him while he is idle but make him some compensation for those anxious and despondent moments which the thought of so precarious a situation must sometimes occasion”. (Smith, 1997, p. 105). The development of estimates of layoff and firm failure risks will allow future researchers to incorporate these variables in a wage equation which tests for any self selection bias.

3. Some behavioral explanations

Able entrepreneurs usually control large organizations. Firms differ in their choice of worker quality partly to economize on monitoring costs. Shirking obviously reduces profits, and a common theme in many models is that it can be deterred by direct supervision or the payment of an efficiency wage. Superior managers realize higher surpluses or rents which they may elect to share with their employees. They adopt a discretionary wage policy where employee compensation is a function of the firm’s performance. These models offer what we call behavioral explanations for the wage–size profile.

3.1. *Monitoring costs and entrepreneurial ability*

McNulty (1980) recognized that there are two human factors of production, labor and entrepreneurs. The latter is characterized by the absence of a market, but the entrepreneur

is the moving force who directs the activities of labor. Neoclassical theory had to reconcile the concept of an optimum firm size with the maintained assumptions of first degree homogeneity of the production function and perfect competition in factor markets. Kaldor resolved this quandary by appealing to the fixity of at least one input.

...it is necessary to assume that the supply of at least one of the factors figuring in the production function should be fixed in which case the "optimum size" ...becomes determinate as a result of the operation of the law of non-proportional returns. Moreover it is necessary that the factor whose supply is "fixed" to the firm should at the same time have a flexible supply to the "industry". In this case, therefore, the fixity of supply must not arise from the natural limitations of the amount available but from a peculiarity of the firm's production function. That is to say, there must be a factor of which the firm cannot have "two" units because only one unit can do the job." (Kaldor, 1934, pp. 66–67)

Management in Kaldor's world involves supervision and coordination. Lucas (1978) argued that individuals with more coordinating ability become entrepreneurs, while the rest are employed by them as workers. Lucas is quite candid when he writes, "This description of management does not say anything about the nature of the tasks performed by managers other than whatever managers do, some do it better than others" (Lucas, 1978, p. 511). Alchian and Demsetz (1972) posit a model where a firm coordinates production and supervises performance. Each entrepreneur in the Oi (1983) model has a fixed endowment of calendar time that is divided between management and supervision. If each worker requires h units of time for supervision, the time available for management is simply $H = (\hat{H} - hN)$. The ability to convert non-supervisory time into efficiency units of management is determined by a parameter λ ; management input $T = \lambda H = \lambda(\hat{H} - hN)$. Output is a function of labor and management inputs, $Q = f(N, T) = f[N, \lambda(\hat{H} - hN)]$.¹⁰ In equilibrium, the marginal value product of labor is equated to its full price, the sum of the wage paid to labor W plus the implicit cost of supervision occasioned by the diversion of time from management to monitoring labor.

$$Pf_N = W + \delta, \quad (5)$$

where $\delta = P\lambda hf_T$. More able entrepreneurs with larger values of λ assemble larger work forces but confront higher implicit monitoring costs. If capital is introduced, greater entrepreneurial ability is accompanied by a higher capital to labor ratio. In an obvious extension, M workers of quality μ can be combined to obtain N efficiency units of labor services, $N = \mu M$. The firm must now maximize profits along both a numbers margin as well as a quality margin. Wages are an increasing function of quality, $W = W(\mu)$ with

¹⁰ One could introduce a capital input K or could introduce labor of different efficiencies. Both extensions are examined by Oi (1983).

$W'(\mu) > 0$. Equilibrium satisfies two conditions:

$$P\mu f_N = W(\mu) + \delta. \quad (6a)$$

$$W'(\mu) = [W(\mu) + \delta]/\mu. \quad (6b)$$

Better entrepreneurs hire more productive workers who entail lower supervisory costs. A positive association between firm size M and wages $W(\mu)$ is generated in this model by matching high- λ entrepreneurs with high- μ employees. The model also implies that the wage structure is convex exhibiting increasing returns to labor quality. If A is twice as productive as B in terms of generating efficiency units of labor services, $\mu_A = 2\mu_B$, then A will earn a wage that is more than twice the wage paid to B because the implicit cost of monitoring an efficiency unit of labor services is inversely related to worker quality. Kruse (1992) found that workers who are more closely monitored are paid lower wages. The ratio of supervisors to workers was higher in large hospitals, Groshen and Krueger (1990). Brown and Medoff (1989) discount the monitoring cost explanation by appealing to the fact that workers on piece rates realize higher hourly earnings at big firms.¹¹ Monitoring is costly, but it is not the principal reason for the firm-size profile.

3.2. Shirking and efficiency wages

The dual labor market model of Doeringer and Piore (1971) implies a wage-size effect. Workers who secure positions in the primary sector are asked to put forth more work effort in return for higher wages, fringe benefits, and job security. A higher wage has to be paid to elicit a larger supply of work effort, $e = e(w)$ with $e'(w) > 0$. The cost of a labor input defined as effort times bodies, $N = eM$, is minimized by setting the elasticity of effort with respect to the wage equal to unity.

$$we'(w)/e = 1. \quad (7a)$$

If $w = w(e)$ is the supply price of effort, the inverse of $e(w)$, cost is at a minimum when the marginal cost of effort is equal to the average cost.

$$w'(e) = w(e)/e. \quad (7b)$$

Many models do not derive the relation of w to e .

Shapiro and Stiglitz (1984) assume that if the cat's away, opportunistic workers will shirk. Such dysfunctional behavior can be discouraged by supervision which raises the probability q of being caught or by imposing a larger penalty for shirking. The wage can be elevated so that a worker enjoys a surplus which will be lost in the event of apprehension. One can determine a "no shirk wage" $\hat{w}$ which has the property that if $w = \hat{w}$, the expected

¹¹ Large employers typically have newer and more expensive equipment which can be more efficiently utilized by hiring workers who are more productive. We amplify on this point in Section 4.3.

utility of shirking (and possibly being caught and discharged) is just equal to the certain utility of working at this efficiency wage $\hat{w}$. The “no shirk wage” will be higher, the larger is the mandated effort standard e , the discount rate r , and the exogenous separation rate b ; it is inversely related to the detection rate q .¹² Instead of an efficiency wage, workers might be required to post performance bonds in advance. This latter option is dismissed by asserting that employers would surely renege. Short sighted employers will allegedly collect the performance bonds and embrace a “hire and fire” policy.¹³ Nothing is said about how a firm sets the effort standard e or how much is spent on setting the surveillance level q . Shapiro and Stiglitz derive an existence theorem, namely unemployment is the penalty which coerces a worker to comply with the mandated effort standard.

A more interesting model was developed by Eaton and White (1983). A firm chooses employment M and an effort e to maximize profits. Monitoring is costly, and workers demand a higher wage premium to supply more effort. There are two equilibria, one where the employee stays on her effort supply curve, and the other where she receives a super-normal wage, the incentive to refrain from shirking.¹⁴ The effort level is now endogenous, and the shape of the effort supply curve determines whether an employer elects to pay an efficiency wage.

The relevance of paying a super-normal wage in large firms to prevent shirking is questioned by Schaffner (1996). In Peru, workers are protected by a no dismissal law. Big employers pay significantly higher wages, but the wage–size premium cannot be attributed to efficiency wages. To the extent that larger employers set higher work effort standards, they must pay higher wages. However, the wage–size premium in this event is a compensating difference discussed in Section 4.1.

3.3. Wages and the ability to pay

Even within a narrowly defined occupation, jobs can differ in tasks, responsibilities, work pace, or location. A high quality worker may be paid a higher wage but may yield a lower “price” of labor. Slichter (1950, p. 80) cautioned that “Neither wages or hourly earnings represent the price of labor”. They are, however, reasonable proxies. A survey of 85 plants in Cleveland in 1947 revealed a very wide dispersion of hourly wages for unskilled labor, from 55 cents to \$1.09 which is at odds with the law of “one price”. Slichter examined the Conference Board data for 20 manufacturing industries in 1939 to find the reason for this dispersion. He discovered several regularities of which at least three deserve mention.

¹² Shapiro and Stiglitz assumed a linear utility function, $U(w, e) = w - e$. Shirking if successful yields a utility gain because e is set equal to zero. However, if caught and discharged, the worker’s wage is replaced by an unemployment stipend. The model yields an equilibrium unemployment rate which serves as a disciplinary device. The same idea is present in the Bulow and Summers (1986) model where unsuccessful shirking puts the worker in the secondary market with its lower wage rate. The limitations of the Shapiro–Stiglitz model are discussed by Oi (1990).

¹³ Shapiro and Stiglitz (1984, p. 442) write, “There is no way to discipline the firm from this type of opportunism”. They ignore the value of establishing and retaining a reputation.

¹⁴ The details of this model are amplified in Oi (1990, pp. S136–S139).

First, the wages of unskilled labor were positively correlated with the wages paid to skilled and semi-skilled employees in the same establishment.¹⁵ Katz and Summers (1989) rediscovered the same regularity. If a firm paid high wages to its skilled machinists and mechanics, it also paid above average wages to its secretaries and janitors. Second, small margins of revenues over payroll were associated with low wages for unskilled labor. Third, firms that enjoy high net income per worker adopt a liberal wage policy.¹⁶

Wages could vary across employers to equalize the net advantages of alternative employments. Tasks, work pace, and fringe benefits are not the same and are responsible for a dispersion of wages. Slichter (1950, p. 91) rejected this hypothesis of compensating differences and embraced a doctrine of discretionary wages. "It reinforces the view that wages within a considerable range reflect managerial discretion. That when management can easily afford to pay high wages, they tend to do so." This idea may have been responsible for what Weiss (1966) called the monopoly-wage hypothesis that workers participate in the excess profits of firms in concentrated industries. Kwoka (1980) tested and rejected this hypothesis using data for blue-collar manufacturing workers from the 1977 Quality of Employment Survey (QES). Although the 4-firm concentration ratio was insignificant, the log of hourly earnings was positively related to establishment size.¹⁷

Hourly earnings across a sample of 41 two-digit industries exhibited a dispersion of 28% which can be reduced to 15% by controlling for worker traits and occupations. Katz and Summers (1989) concluded that the inter-industry wage structure was stable over time.¹⁸ They contended that the inter-industry wage differences are not compensating differences but are the results of sharing rents. Boiler makers may get premiums for their exposure to noise, but why does the employer pay a premium to the secretaries who write out their paychecks? An alternative to the rent-sharing hypothesis is that high-wage industries employ a higher quality of labor where quality is unobservable. Katz and Summers ask the rhetorical question, "If sorting does not take place on observable traits such as education, why should a firm sort on an unobservable attribute?"¹⁹

Katz and Summers try to save the thesis that industry wage differentials represent labor

¹⁵ The rank correlation coefficient across 20 industries was +0.7098. The data were collected by the National Industrial Conference Board, *The Conference Board Economic Record*, Vol. II, No. 10 (March 28, 1940), pp. 120–134.

¹⁶ The rank correlation between net income per employee and the hourly wage of unskilled labor was +0.6969. Slichter chose to compute bivariate rank correlations for some nine hypotheses. He did not estimate any multivariate relations.

¹⁷ In his regression (c), Kwoka (1980, p. 370) obtained a significant coefficient on the 4-firm concentration ratio, but when he added a vector of industry dummies, the partial effect of concentration became statistically insignificant. Weiss (1966) also rejected the monopoly-wage hypothesis.

¹⁸ The correlation of relative wages in 1966 and 1984 across 41 industries was +0.91. Schultze (1989, p. 281) questioned the so-called stability claimed by Katz and Summers. He noted that for the seven high wage industries (Mining, Transport Equipment, Tobacco, Petroleum, Communications, Public Utilities, and Primary Metals), the average of the raw wage premiums was 32% in 1966 and 50% in 1980.

¹⁹ The bivariate tabulations shown in Table 4 reveal that employees in large firms are, on average, of higher quality judged by education, job tenure, and marital status of men. The evidence is less clear on sorting across two-digit industries.

rents by appealing to evidence on quit and vacancy rates. Based on data for 74 two-digit manufacturing industries, the correlation between wages and quit rates was -0.70 . Additionally, there were 14.0 applicants for each job vacancy in the five high-wage industries compared to only 8.5 applicants in the five low-wage industries. These findings persuade the authors to conclude that the inter-industry wage structure is generated by labor rents. High wages incorporate labor rents which are sustained by the discretionary benevolence of wealthier employers or by the need to pay efficiency wages.²⁰ The surplus of revenues over labor costs varies and is probably relatively larger in bigger firms which face lower prices for non-labor inputs and may have market power. The incidence of unionism is positively related to firm size. Trade unions capture a part of these surpluses in the wage premiums which they secure for their members. Threat effects may raise wages in large non-unionized firms, though Brown and Medoff (1989) found little support for this possibility. Brown et al. (1990) document the relation between wages and per worker surpluses, but they are at a loss to rationalize why employers want to share these rents with labor.

4. The productivity hypothesis

The investigation upon which we are about to enter as to the influence of the status of the laborer of the concentration of industry in large establishments is both of theoretical and practical importance. Its practical value lies in the answer to the question as to whether the form of selection of laborers entailed by the survival in competition of large establishments places the employees upon a better plane of living than the one occupied by their fellow workers in smaller establishments. Its theoretical interest lies in the answer to the query as to whether the productivity hypothesis will explain the results to which the investigation will lead. (Moore, 1911, p. 139)

That wages are positively related to the size of the establishment and the firm is a well-established fact. In the preceding section, attention was directed to three “behavioral” explanations: (1) matching workers and entrepreneurs to minimize the sum of wage and monitoring costs; (2) paying efficiency wages to deter shirking; and (3) sharing the surplus of revenues over labor costs with workers. A strategy followed by some is to look for the right variable or vector of variables which when included in a wage equation, reduce the partial effect of firm size to zero. The covariates of firm size are numerous, and new surveys keep washing up additional covariates. Henry Moore was guided in his search for variables that could confirm or reject a productivity hypothesis.

²⁰ They appeal to the Akerlof (1984) gift exchange. If a worker is paid a fair wage, she will put forth more effort. Raff and Summers (1987) contend that Ford paid a super-normal wage as a way of buying peace and avoiding the kind of shirking found in other industrial settings. Further, Ford wages were raised when assembly line technology was introduced to improve discipline and to cut down on absenteeism. Again, this looks like paying for more work effort, a compensating difference and not an economic rent. The estimates of labor and capital rents can be found in Table 11 of Katz and Summers (1989).

We begin with a maintained hypothesis, namely wages and productivity move together. A labor contract is almost always explicit about the rate of wages but is vague about “the effort bargain”, what constitutes a fair day’s work. An effort standard is an unobserved condition of work. Raising that standard results in greater productivity which can support a higher wage, a topic discussed in Section 4.1. The pace of work depends, in part, on customer arrival rates in the distributive trades and the service sector. The wages in bigger stores and hospitals are higher because workers have to work harder (Section 4.2). We turn in Section 4.3 to the complementarity of capital and skilled labor. The analysis is complicated because both capital and labor are heterogeneous. Firms try to match their most powerful machines with their “best” workers where ability is often unobservable. The volume production of standardized goods is most efficiently organized around teams. Section 4.4 examines how conformance with a common set of work rules and practices can be efficient but requires the payment of “rents” to infra-marginal team members. The closing section deals with the decision by larger firms to “produce” more productive and durable employees.

4.1. Effort, productivity, and the disutility of work

A job involves more than an exchange of money for time. Some dimensions of the job package (such as the number of paid vacation and personal leave days, health and life insurance) can be explicitly incorporated into a labor contract. Compliance with autocratic orders, exposure to an unhealthy workplace, or long commutes are some negative job attributes. Positive aspects include challenging tasks, opportunities to talk to co-workers, exercising initiative, or taking time off for personal business. Large employers typically provide cleaner and safer workplaces, better lighting and climate control, generous time-off benefits, and superior fringe benefits when compared to small employers. Working conditions, at least the observable ones, are superior at larger firms and hence cannot explain higher wages.

Employment agreements, written or implicit, rarely stipulate the effort that the employee is expected to supply.

Central to the worktime issue is the concept of a fair day’s work....The differing perspective (between employer and worker) is not resolved by the forging of an employment contract. Such an arrangement normally consists of two elements: first, an agreement on the wage per unit of time or piece; second, an agreement on the amount of work to be undertaken, that is, an effort bargain. It is normal for the wage rate to be precisely defined in the employment contract. The effort bargain, on the other hand, is generally implicit and indistinct (Nyland, 1989, p. 57) ...the formal wage contract is never precise in stipulating how much effort is expected for a given wage (and vice versa). The details of the arrangement are left to be worked out through the direct interaction between the partners of the contract. (Baldamus, 1959, p. 35)

Disagreements about the duties and responsibilities of the two parties can affect morale, absenteeism, and may even lead to a job separation. If effort bargains are so important, why are they left vague and open ended? Simon (1947) offered two reasons: (a) Nothing will be gained by the organization without the employee's acceptance of the authority. (b) The precise activities that he performs within broad limits are indifferent to him. Contracts are meant to facilitate mutually advantageous exchanges, to expedite the allocation of resources to their highest valued uses. A contract that tries to anticipate too many contingencies becomes unwieldy and raises transaction costs. Some things like effort, responsibility, and loyalty might better be left understood.

Labor productivity varies not only over the hours of a day but also over the days of the week. The weekly profile almost always follows practice-efficiency. Output is lowest on Monday, rises to a peak on Wednesday or Thursday, and falls off on Friday. Vernon (1921, pp. 27–28) wrote:

The cessation of work between Saturday afternoon and Monday morning naturally causes a greater loss of neuro-muscular co-ordination than that observed between each week day, and consequently the output on Monday morning tends to be lower than that observed on any other morning of the week. The loss of practice-efficiency owing to the week-end rest is so considerable that the remainder of the week may be needed for recovery, but...the fatigue induced by the daily round of labor gradually accumulates...first to neutralize the improvement due to practice-efficiency and then to overpower it.²¹

The model is one in which output is positively related to a stock of practice-efficiency capital P and current effort E . At the start of a week, the stock of P -capital is especially small resulting in the Monday effect. Although P -capital is replenished by work, it depreciates with the passage of time. The accumulation of work de-energizes a worker to the point where he can supply less current effort E . The latter effect overtakes the build-up of P -capital producing the Friday dip. Finally, the supply of energy is not fixed. It can be augmented through rest and nutrition. Vernon reported that piece rate workers took spontaneous breaks averaging eight minutes an hour. Output is a function of effort, but the potential supply of effort is not fixed. Jobs differ in their effort intensity, but so also do non-market activities. Watching television requires less effort than caring for a child. Becker (1985) conjectured that women might select less effort intensive and hence lower paying jobs than men because they want to conserve more effort for non-market activities.

In addition to the rate of pay, the nature of the work surely influences the amount of time and effort supplied to a job. Sebastian de Grazia (1962, p. 58) described the situation as follows:

We should not expect the worker normally to get real satisfaction from a job on

²¹ He continued, "In industry, it is almost invariably accompanied by low output, which Kent has termed, the Monday effect." (Kent, 1915).

which he does a piece of the work, a task chosen and organized by others, under watchful eyes, at a pace not his own, at a time and place not his to say. This, one would venture, is the disagreeable part of the job effort or exertion or exercise, physical or mental, under orders and supervision, constrained in time and space.

Physically or mentally demanding work is one dimension of a job. Baldamus reserved the term, boredom, for professional positions and introduced the concept of tedium for dull, repetitive jobs. The disutility of the latter kind of work can be partially offset by traction, the opposite of distraction. "It is the feeling of being pulled along by the inertia inherent in a particular activity. The experience is pleasant and may therefore function as a relief from tedium." (Baldamus, 1959, p. 69). At a confectionery, wrapping can acquire a rhythm and hence traction, weighing cannot. Rhythm and traction go together to reduce the disutility of a tedious job. Putting several tasks into a job might make it more attractive, but Baldamus (p. 60) found that frequent changes from one operation to another interfered with the swing of work resulting in lower output.

Work in the primary sector is characterized by strict work rules, performance standards, and incentives to supply more effort. A higher effort level means a larger flow of labor "services" measured in efficiency units. Wages rise with firm size, but labor productivity climbs even faster resulting in lower unit labor costs. Kwoka (1980, pp. 363–366) asked if differences in the organization of work led to greater "alienation" of employees in larger firms? Using a variety of subjective measures of job satisfaction, he concluded that "worker alienation" was unrelated to establishment size.²² The payment of higher wages evidently compensated them for the stricter work rules and effort standards. In short, the observed wage–size profile can be supported by higher effort levels that result in greater labor productivity.

4.2. *Economies of massed reserves*

The production function applicable to firms in the distributive trades and service industries differs from that for firms making goods in two important respects (Oi, 1992). First, the customer supplies an essential input. The number of customers/buyers B per unit time period belongs as an argument of the production function alongside labor and capital, $X = f(L, K, B)$. Second, customer arrivals are random resulting in a stochastic production function. Patient care is produced by bringing together nurses L , beds K , and most importantly, patient-buyers B . A retail store produces transactions by combining customers, clerks, inventories, and capital facilities. Without a customer, a retailer could not produce a transaction, the *raison d'être* for its existence. A store assembles an inventory, hires sales clerks, and stays open even when there are no customers. A teddy bear sits on a shelf awaiting the arrival of a buyer. On the other side, a shopper travels to a store and sometimes has to wait to be served. Something or someone is almost always hanging around.

²² Brown et al. (1990, p. 36) identified 42 working conditions of which 21 were marginally worse in large establishments. An adjustment for the quantifiable variables appears to magnify the size–wage premium.

Hutt (1977) is correct, idle resources are productive when they are in a state of pseudo idleness. All idleness could be eliminated, but to accomplish this, the synchronization of customers, clerks, and just-in-time inventories would be prohibitively expensive. Idleness is also observed in manufacturing. Factories are closed at night, workers get rest breaks, and inventories are held. Neglecting these considerations apparently poses no serious theoretical or empirical difficulties in analyzing the goods producing sector. This is not so for the service sector.

It is instructive to review the repairman's problem in queuing theory. A firm has M machines. The number that arrive to be repaired follows a Poisson distribution with a mean arrival rate λ . The repair time is exponentially distributed with a mean service time μ . With only one repairman, there is some probability p_0 that none of the machines needs servicing, and the repairman is idle. When two or more machines break down, a queue develops. The addition of a second repairman raises labor costs but reduces the opportunity costs of idle, broken machines yielding no output. If λ and μ are known, we can solve for an optimum ratio of machines to repairmen which minimizes the sum of costs for labor and idle machines. The problem is isomorphic to the problem facing a hospital serving a population of M potential patients who arrive at random to be treated by R "servers". With a mean patient arrival rate λ and a mean length of hospital stay μ , the sum of waiting times (empty beds R awaiting a patient and patients waiting for a bed to become available) is minimized when the ratio of the population to beds satisfies the equation, $(M/R) = [1 + (\lambda/\mu)]^{-1}$. If M and R are both doubled, the mean length of a patient queue will fall and the occupancy rate of beds will rise.²³

A firm in the service sector confronts a production function exhibiting increasing returns which is a consequence of the economies of massed reserves. Robinson (1958) showed that these economies derive from the coordination and synchronization of activities which are related to the scale of operation and are not a result of the law of large numbers. The implications of these economies are borne out in the data. A queuing model implies that if half of the hospital bed capacity in a community is controlled by the largest hospital, that hospital will realize more than half of the occupied beds. The same principle applies to airlines. If two or more carriers serve a given route, the one supplying the largest number of available seats achieves the highest load factor defined as the seat occupancy rate. A passenger-trip, a hotel room, or an exchange of goods for money may constitute the main component of the "product" of a service establishment. An examination of selected industries reveals that the output mix varies with firm and establishment size.

The corner grocery store and the giant supermarket belong to the same three-digit industry, SIC 541. However, they differ in ways that affect the relation of outputs to inputs. We begin with the capital to labor ratio. The ratio of the book value of capital to employment is higher for larger stores, but selling area and inventories per employee are lower. Delicatessens, fresh fish markets, and bakeries use less floor space but more capital equipment. Buildings and equipment are newer in bigger stores which make more use of

²³ The mathematical model is nicely summarized by Mulligan (1983).

scanners.²⁴ The capital/labor ratio is higher in bigger stores when the data are adjusted for differences in the output mix and the age and quality of capital. Larger firms utilize capital more intensively, by operating more shifts per day in manufacturing and by staying open longer hours per week in retailing. The smallest independent supermarkets were open an average of only 91 h a week in 1994 compared to 135 h for the largest supers (see Table 11).²⁵ The rise in the proportion of two-earner families and the spread of car ownership rates are responsible for the changing structure of retail foodstores. Higher capital/labor ratios and longer store hours have raised labor productivity, but these have not raised the real wages of retail workers.

A second significant development is the change in the relation of customer to vendor. In 1950, one could still be served by a butcher or retail clerk. Self service, an institutional innovation started in the depression, is now the rule, even at convenience stores. National advertising and brand names have replaced trained clerks who advised customers and honored implicit warranties. They have undermined the value of store specific human capital. Transactions have become impersonal and standardized. There is little to be gained from establishing on-going relations between customers and trained clerks who know one another. Bluestone (1989) concluded that we have witnessed a *Retail Revolution* and its byproduct, a de-skilling of the retail workforce. Supermarkets have increased their demands for part-time employees. Full-time clerks outnumbered part-timers in 1975, but by 1988, full-timers made up only 41% of workers at the independent supermarkets and 36% at the chain supers. Store size and the percentage of part-time employees are positively correlated (see the last column of Table 11). This pattern emerges even though the relative wages of part-time workers are higher at the larger supermarkets. Differences in the staffing practices of large and small supermarkets can be explained by a peak load model. A part-time clerk is like a standby generator which comes on-line only at peak periods. The base demand in peak and off-peak periods is satisfied by year-round generators, the counterpart of full-time employees. The ratio of standby to year-round generators will be higher, the greater the gap between peak and base demands. The giant supers stay open longer and find that fully 55% of sales take place in 3 days, Thursday to Saturday. In relation to small supers, the big markets confront higher and longer peak sales periods which are staffed with part-time workers. The de-skilling of the retail work force over the last quarter century is due in large measure to the increasing dispersion of hourly sales.

Table 12 confirms the presence of the size-wage premium. Wages of part-timers at the largest independents were 18.3% higher than at the smallest. The premium was 16.3% for full-time clerks.²⁶ Customer arrival rates, the size of the market basket, and sales volume are all greater at larger supers. Labor is more productive as evidenced by sales per

²⁴ In 1981, the average age of buildings occupied by the largest supers was 9.8 years versus 16.6 years for small supers; see *Progressive Grocer* (April 1982, p. 23). The percentage of stores using scanners is positively related to size.

²⁵ Only 2% of independent supers with less than 10,000 feet² of selling area were open 24 h, 7 days a week. The fraction climbed to 47% for the largest size group. The mean store hours and percentage staying open all of the time were higher for the chain stores.

Table 11
Selected statistics by store size, 1994 (independent supermarkets by selling area)^a

Selling area	Weekly sales (000)	Sales per employee hour	Basket size	Employees total	Percent part-time
<10	67.0	76.46	11.36	30.1	53.8
10–15	84.5	80.32	12.66	37.3	56.8
15–20	123.3	80.11	14.19	54.3	55.3
20–25	162.1	87.61	15.79	67.1	60.1
25–30	182.8	83.97	16.33	77.5	61.3
30–44	302.6	85.86	19.88	117.8	60.0
40+	446.5	109.80	22.99	135.9	63.9
Average	127.1	83.86	14.81	52.7	57.9

^a Source: *Progressive Grocer* (April 1995) Supplement, pp. 47–52.

employee hour shown in Table 11. An employee at a larger store has to work harder. She has less idle time, accepts more responsibility, and receives a higher wage. A firm gets larger because it can attract more customers who supply inputs into the production of the services supplied by the firm. A higher customer arrival rate leads to more efficiency which supports the higher wages of more productive workers due to the economies of massed reserves.

4.3. Capital, technology, and labor productivity

Productivity and wages depend on the amounts and kinds of cooperating inputs provided by the employer. The capital to labor ratio is positively related to firm size. Large manufacturing establishments utilize capital more intensively by operating multiple shifts, a fact documented by Foss (1981). According to Shinohara (1962), used machines and equipment account for fully 40% of the assets of small Japanese firms, while large firms turned to the used markets for only 6% of their capital acquisitions.²⁷ The higher relative demand for capital, especially new equipment, can be explained in part by the fact that larger and older firms confront lower “prices” for non-labor inputs.²⁸ Interest rates are lower, and sellers of advertising, energy, communication services, and insurance typically offer

²⁶ Wages pertain to 1985, and size is measured by sales volume. The size–wage premium for the chain stores were larger, 37.8% for part-time and 22.8% for full-time clerks.

²⁷ Firms in the United States behave in the same manner. A Survey of Truck Inventories and Use contains data on the age distribution of truck-tractor fleets. Eric Bond at Pennsylvania State University supplied us with tabulations dividing fleets by size into thirds. The median age of trucks in the fleet was 6.3 years for small fleets and 4.8 years in the largest fleets.

²⁸ Evidence for this proposition is supplied in Chapter 4, especially Table 4.8 of Brown et al. (1990).

Table 12
Hourly wages (\$) of part-time and full-time clerks: 1985^a

Sales volume (in millions of dollars)	Part-time clerks	Full-time clerks	Ratio, part-time/ full-time
<i>Independents</i>			
2-4	3.98	4.84	0.822
4-8	4.31	5.11	0.843
8-12	4.64	5.35	0.867
>12	4.71	5.63	0.837
Average	4.18	5.01	0.834
<i>Chains</i>			
2-4	4.15	5.45	0.761
4-8	5.07	6.23	0.814
8-12	5.49	6.45	0.851
>12	5.72	6.69	0.855
Average	5.10	6.23	0.819

^a Source: *Progressive Grocer* (April, 1986).

volume discounts. Finally, the largest and most profitable firms are usually the first to adopt new product and process innovations. The task at hand is to explore how these empirical regularities affect wages.

4.3.1. A neoclassical production function

Across the size spectrum, firms embrace different technologies and factor proportions. Assume for the moment that firms use the same production function combining capital K , skilled labor A , and unskilled labor B ; $X = f(K, A, B)$. Large firms face a lower price for capital which is substituted for labor. The capital to output ratio (K/X) as well as the capital to labor ratio (K/L) are higher in larger establishments. A lower input price not only expands the demand for capital K but also for its complement, skilled labor A . Hamermesh (1993) and Griliches (1969) found that capital and skilled labor are complements in production. A larger firm will demand a richer skill mix (A/B) and pay a higher average wage. But this is not what we mean by a size-wage premium. An inspector at a big bottling plant earns more than an inspector at a small brewery even though the latter could, in principle, perform the tasks of the former. What sustains the size-wage premium? The quality of the machinery and/or the work pace might differ across the two bottling plants. Hence, the "work" performed by the two inspectors is not the same.

The theory has to be embellished. At a minimum, why do large firms embrace a more capital intensive mode of production? One candidate is a non-homothetic production function that yields an output elasticity of demand for capital that exceeds unity. A second proposed by Brown et al. (1990) appeals to input price differences. An airline that orders

twenty new jet aircraft enjoys a lower unit price than a competitor who demands only one or two planes. The availability and terms of credit are more favorable for larger firms. The supply price of capital has to be firm specific, a function of the firm's age and size, its credit rating, reputation, and the collateral it can put up. A lower relative price for capital can explain higher relative demands for both capital and its complement, skilled labor. But why does the large employer have to pay more for skilled workers? A competitive market paradigm implies that the wage of an inspector ought to be independent of the size of the bottling plant. Yet wages are higher in the big plant. Adam Smith told us that competition equalizes the net advantages of competing employments, not wages. There must be some other factor that differs across plants. A higher ratio of machines to workers may produce more noise. The installation of new machines might prompt management to step up the pace of the assembly line. The acquisition of capital will increase the demand for a complementary input, skilled labor, but it is often associated with a faster work pace raising labor productivity and hence wages.

4.3.2. *Wage premiums for computer operators*

Technological advances introduce new machines and new ways of doing things that frequently increase the demands for cooperating inputs. Many of the innovations adopted by manufacturing plants involve the use of computers. Numerically controlled machines led the way, but the major changes followed the entry of the microprocessors, the Apple-2 in 1977 and the IBM-PC in 1981. Autor et al. (1997) report that the percentage of workers using a computer rose from 25.1% in 1984 to 46.6% in 1993.²⁹ Krueger (1993) estimated that in 1984, a worker who used a computer was paid an hourly wage that was 0.17 log points greater than a comparable worker who did not; the gap rose to 0.20 log points in 1988. His data regrettably did not identify firm or establishment size. A 1979 survey of Canadian workers in the Maritime Provinces inquired if the employee had access to a computer. Reilly (1995) estimated a wage equation in which the coefficient of establishment size was positive and significant when computer presence was excluded. A computer raised wages by 0.13 log points, but the effect of size fell to nearly zero.³⁰ Reilly concluded that computer access solved the mystery of a size-wage premium but replaced it with another, namely why should computer technology lead to such a large enhancement in a worker's stock of human capital?

²⁹ The direct use of computers at work varied by worker characteristics. Data from the 1993 CPS supplement (reported in Autor et al., 1997, Table 4) revealed computer use rates for men 41.1, female 53.2, less than high school 10.4, high school 34.6, some college 53.1, college 70.2, white 48.0, black 36.7, blue collar 17.0, white collar 67.6, union 39.1, non-union 46.9, part-time 29.3, full-time 51.7%.

³⁰ Reilly's sample contained 607 workers in 60 private sector establishments in manufacturing and non-manufacturing industries. Data from the General Segmentation Survey were obtained on employment size, the ratio of supervisors to employees, capital per employee, education requirements on new hires, dummy on average age of capital, and access to a computer as well as the usual worker traits. The coefficient of size was +0.035 with no other covariates, +0.039 including establishment characteristics other than computer access, and +0.011 with all covariates. Computer presence in this sample obviously exhibited a high colinearity with establishment size.

Table 13
Technology use and education of the workforce, 1988–1990^a

In plants with <i>X</i> technologies	% of all workers with a college degree	% of non-productive workers with a college degree	% of production workers with some college
<i>X</i> < 4	9.4	24.1	21.2
<i>X</i> = 4–6	12.2	31.2	24.2
<i>X</i> = 7–8	14.0	34.5	27.1
<i>X</i> = 9–10	16.2	34.9	27.7
<i>X</i> = 11–13	15.2	37.5	29.7
<i>X</i> = >13	33.1	53.9	34.9
Full sample	18.3	40.1	27.9

^a Source: Doms et al. (1997, Table 1, p. 262).

A 1988 survey collected data on the adoption of new technologies.³¹ For a sample of single product plants, Dunne and Schmitz (1995) found that the incidence of establishments adopting six or more new technologies rose from 7.9% for plants with less than 100 employees to 85.4% for those with 500 or more workers.³² Taking the plant as the unit of analysis, Dunne and Schmitz regressed the log of the hourly wage of production workers on plant size, technology use, and other characteristics of the worksite. Wages were positively related to both size and technology use. The introduction of technology use attenuated the effect of employment size on wages, but size was still significant.

A strong correlation between technology adoption and the skill mix is evident in Table 13.³³ The percentage of employees with a college degree ranged from 9.4% in plants with fewer than 4 new technologies to 33.1% in highly technical plants. The skill upgrading for blue-collar workers, shown in the third column, is less than that for white-collar workers. Controlling for education reduces the partial effect of plant size, but size is still important.

Technical progress in the last 25 years has embodied a strong skill bias. Investments in new equipment, especially computers, were accompanied by sharp increases in the

³¹ The Survey of Manufacturing Technologies collected data from 10,526 plants with 20 or more employees in five two-digit manufacturing industries; Fabricated Metals, Machinery, Electrical Machinery, Transportation Equipment, and Instruments. The survey collected information on if the establishment used any of 17 advanced technologies including such things as computer aided design, flexible machines using programmable controls, pick/place robots, technical data network, etc. The survey did not ask about the fraction of workers who actually used the technologies in question.

³² The figures for the intervening size classes were 24.9% in the 100–250 size class, 63.3% in the 251–499 class and 24.0% for the entire sample of 1837 single product plants. The data were taken from Table 2 of Dunne and Schmitz (1995).

³³ The number of advanced technologies was obtained from the 1988 Survey of Manufacturing Technologies. Education and payroll data come from Troske's Worker-Establishment Characteristics Database, a matched sample of 34,034 workers and 358 plants (Troske, 1999).

demand for workers with more formal education and work experience whose wages rose relative to the wages of less skilled employees.³⁴ The firms adopting the new production methods happen to be larger and older because there are major overhead costs. The story is that a big firm adopts a computer based production method which calls for highly skilled workers. Doms et al. (1997) assemble some data supporting an opposing scenario wherein skilled labor is the fixed factor. Older and larger plants happened to have assembled a work force of more skilled employees and were hence better situated to adopt the new innovations. The larger dataset analyzed by Autor et al. (1997) tend to support the first story in which skilled labor is a variable input. More research is required to resolve this issue.

4.3.3. Capital and labor productivity

Capital is not an amorphous, homogenous input. Capital imbedded in an air conditioning system or better lighting is a local public good. It raises the marginal product of a collective group. Little is to be gained by trying to estimate the impact of a local public good on the productivity of any particular group of employees. Land exemplifies a capital input that can be divided and allocated to workers. If land is homogenous, and labor is not, a firm will maximize output by allocating more land to more productive individuals. Further, if land is of uneven productivity, the most productive plots will be assigned to the best workers. Given heterogeneous inputs, the capital/labor ratio will not be equalized across employees in the same establishment. Given differentiated capital goods, a firm faces an assignment problem. A new computer will be allocated to the most efficient operator who has the highest shadow price. The older and less powerful calculators and word processors are passed down to less productive employees. This assortative mating of equipment and employees results in a maximum of output. The problem is analogous to a monogamous marriage market where Becker (1973) argued that “output” is maximized through the assortative mating of above average men with superior women. Sorting and matching of equipment and employees of varying qualities take place within and across establishments. Moore (1911, p. 148) observed:

Because of the use of large fixed capital in large establishments, the more efficient workers are more valuable to the large than to the small establishments.

Moore (p. 149) goes on to quote from Marshall:

We have hitherto supposed that it is a matter of indifference to the employer whether he employs few or many people to do a piece of work, provided his total wages-bill for the work is the same. But that is not the case. Those workers who earn most in a week when paid at a given rate for their work are those who are cheapest to their employer.... For they use only the same amount of fixed capital as their slower fellow workers; and, since they turn out more work, each part of it has to bear a less

³⁴ These trends are discussed by, among others, Behrman et al. (1994), Dunne and Schmitz (1995), and Autor et al. (1997).

charge on this account. The prime costs are equal in the two cases; but the total cost of that done by those who are more efficient, and get the higher time-wages, is lower than the total cost of that done by those who get the lower time-wages at the same rate of piece-work payment.... But when expensive equipment is used which has to be proportioned to the number of workers, the employer would often find that the total cost of his goods lowered if he could get twenty men to turn out for a wages-bill of £50 as much work as he previously got done by thirty men for a wages-bill of £40.

Hourly wages are positively related to equipment value. The occupational compensation survey for Seattle reported the following wages for truck drivers, \$9.45 for light, \$11.15 for medium, \$14.68 for heavy trucks, and \$15.74 per hour for tractor-trailers.³⁵ Annual earnings rise from around \$30,000 to \$100,000 when a pilot moves from flying a small plane for a commuter airline to a large jet for a domestic trunk airline. Imagine an equation in which the log of hourly earnings is regressed on, among other things, a dummy variable equal to one if the driver operates a tractor-trailer and zero otherwise. The coefficient will undoubtedly be positive and might be interpreted in several ways. The acquisition of tractor-trailers increased the demand for more skilled drivers, a capital/skill labor complementarity hypothesis. The larger vehicle raises productivity of the unit and yields a larger surplus which is shared, an ability to pay or what some call a rent-sharing story. Driving a larger truck or flying a bigger jet means the acceptance of more responsibility and the assumption of more effort to protect the larger investment in equipment and cargo. This last suggests that the coefficient of a dummy variable for tractor-trailer represents a compensating difference.

Large establishments surely attract and retain more capable workers. Moore (1911, p. 163) claimed that:

The large establishments are able to carry out the work of selection (of more capable individuals) because, in consequence of their large capital and better organization, they offer opportunities for more capable laborers to reap the reward of their differential ability.

Labor productivity can vary across individuals and firms as well as over time. Some people are inherently more productive, an unobservable person-specific trait. An employer can supply the worker with a larger quantity of a cooperating input or access to a superior technology. She can also demand more work effort or insist upon a stricter performance standard which looks like an unobservable job-specific characteristic. Larger firms are better informed and have access to more favorable credit terms to acquire new capital goods. The returns to capital and adoption of new process innovations are enhanced by raising performance standards. Advanced technologies, inherently superior employees,

³⁵ Communication from Professor Michael H. Belzer, University of Michigan, May 28, 1997. The data pertain to December, 1992.

and higher work standards go together to raise labor productivity which supports the relation of wages to size.

4.4. *The organization of production*

An industry is defined as a set of firms that produce the same product. A firm assembles inputs, allocates them to different activities, coordinates and supervises them, and is the residual claimant to the surplus of revenues over costs, be it positive or negative. The size of a firm depends on a host of variables. Technology clearly matters. Barber shops are small, many have no paid employees. Mobile homes are manufactured by relatively small firms because of the high costs of transporting the final product. The technical economies of scale encourage the survival of large utility companies. Even a narrowly defined industry exhibits some product differentiation. The smallest firms in “Contract Construction” make home repairs, remodel kitchens, and convert garages into family rooms. High-rise apartment and office buildings are built by large companies. Big firms in the fabricated aluminum products industry manufacture standardized goods (window frames and siding) which are produced in capital intensive assembly line factories and are marketed through a nationwide distribution network. Customized aluminum extrusions (machine tools and gutters) are usually supplied by small firms. Giant super stores and convenience grocery stores belong to the same industry, SIC 541, but offer different product lines and services. Access to capital is an important determinant of size and product type. Small firms confront a higher price for capital and usually embrace older vintages of technology. Little et al. (1987) found that small soap factories did not have sufficient wealth to purchase electricity and produced a lower quality product. A fall in the price of electricity ought to lead to larger soap factories.

Chandler (1982) observed that the success of many giant firms can be traced to the introduction of a good or service which was produced in large volume. The rate or flow of output per unit time period X is the usual metric by which firm size is measured in our received theory of price. Alchian (1959) developed a model in which the rate X and accumulated volume V are two components of an output program. Production is characterized by increasing returns in the sense that the unit cost (in present value terms) is a decreasing function of volume.³⁶ The economies of volume production are achieved by producing a standardized good via batch, assembly line methods. The firm incurs high overhead costs for specialized equipment that is amortized by utilizing it more intensively. Foss (1981) found that only 5% of blue-collar workers in small manufacturing plants were assigned to evening or night shifts compared to 23% in large plants. Large organizations

³⁶ An output program is described by the rate of output per period X , the accumulated volume V , and the planned interval over which the output will be produced. The several dimensions are not mutually independent. Cost is an increasing function of the output rate X holding constant the total planned volume. An increase in volume holding the output rate constant leads to a decrease in the unit cost (C/V). The Alchian model is based on the cost analysis of aircraft production by Asher (1956) and is implicit in the early article by Wright (1936). The connection to production functions and learning is discussed by Oi (1967).

are more cheaply administered by standardizing products and processes, a fact recognized by Clark (1923, p. 96) who opined, "... and the satirist of *Main Street*, Mr. Sinclair Lewis, is doing his best to convince us that business is producing standardized people".

Consider two imaginary breweries. B1 has a product that has caught on and is capturing a large national market, while B2 supplies a local market where retailers handle several brands. A disruption in output is costlier for B1 who has established a marketing network. His distributors and wholesalers depend on a steady flow of output. The customers of the local brewery B2 incur a lower cost from a breakdown in output. B1 will acquire newer equipment and invest in a safer workplace. Injury rates are a lot lower in bigger plants as shown by Oi (1974). A steadier flow and larger volume of output enable the larger brewery B1 to maintain a lower inventory to sales ratio. The inspectors in the bottling plant and the men on the loading docks work harder because there are fewer breakdowns and there is less idle time due to the economies of massed reserves. Hall (1991) offers another example. The productivity of a person making sandwiches in Manhattan is an order of magnitude greater than that of a sandwich maker in a small town. Large employers who engage in the volume production of standardized goods and services demand more reliable employees and ask them to put forth more work effort. These employees earn higher wages than their friends who work for the small brewery with its slower work pace and smaller volume.

Every evening, a hundred women leaned over a hundred fires to prepare a hundred inedible meals. Better meals at lower costs could be obtained by giving the job to a team, a chef and his two assistants.³⁷ Team production is efficient. A person who only straightens and points the wire in Adam Smith's pin factory can learn to do this well. Time can be lost and the rhythm of work broken if one is asked to shift from one task to another. Increasing returns often accompany cooperative effort when two or more men join together to pull a barge up a canal, assemble trucks, or mine copper. Workers are placed into teams even though it becomes harder to ascertain the productivity of individual team members.

In "The O Ring Theory of Economic Development", Kremer, 1993 assumed that output Y is a multiplicative function of capital and the labor inputs of N team members.

$$Y = AK^\alpha [Hq_j]^\beta = AK^\alpha [q_1 \times q_2 \times q_3 \times \dots \times q_N]^\beta. \quad (8)$$

The input by the j th employee represents her productivity as a fraction of a perfect employee so that $0 < q_j < 1$ for all j . The social product can be raised by segregating workers, putting high- q workers on one team and low- q individuals on another. High- q secretaries are matched with high- q lawyers. Low- q persons earn lower wages but can compete with high- q workers because their respective "teams" produce differentiated products. Given team size, a firm's output is an increasing function of the mean quality of its work force and a decreasing function of the dispersion of worker qualities. Team production is more efficient when workers are alike and perform tasks in the same way. By

³⁷ This is one example of the inefficiency of the family described by the 19th century utopian, Charles Fourier whose life and works were chronicled by Beecher (1986).

Table 14

The distribution of team size by size of establishment (Quality of Employment Survey, 1973)^a

Establishment size	1-9	10-49	50-99	100-499	500-999	1000-1999	2000+
	(131)	(172)	(69)	(120)	(49)	(31)	(61)
Team size	3.22	7.54	9.36	10.66	10.94	12.90	14.31
	(3.63)	(7.77)	(12.13)	(10.75)	(16.59)	(18.24)	(15.78)

^a Source: Idson (1995, Table 2, p. 199). *Note:* Cell sizes are reported in parentheses below the plant size category headings. In each cell the mean team size is reported with the standard deviation in parentheses.

allowing β to depend on team size, Kremer can generate a function exhibiting increasing returns. Larger teams recruit higher quality workers who are combined with high-valued capital inputs.

How prevalent are teams? In a sample of 633 non-union workers responding to the 1973 Quality of Employment Survey, 14.4% did not belong to a team. The rest belonged to teams whose average size was 8.6 persons.³⁸ Table 14 confirms the anticipated correlation between establishment and team sizes. Idson (1995) found that a 1% increase in team size was associated with a 3.5–4% increase in annual earnings. Deardorf and Stafford (1976) explain this outcome from a theory of compensating differences. If team members are obliged to work in tandem, they must agree on several dimensions of the job, the length of the workweek, work pace, number of rest breaks, indoor temperature, and so on. At an hourly wage of $w_0 = \$10$, workers A and B might want to supply $H_A = 35$ and $H_B = 45$ h a week. If A is required to supply 40 h, he will require a compensating earnings difference of more than $5 \times 10 = 50$ dollars to stay on the same indifference curve. Likewise if B is forced to reduce her weekly hours from 45 to 40, she will accept a cut in earnings but by an amount less than $w_0(H_B - H^*) = 50$ dollars. The common wage-hours package, (w^*, H^*) assumes the character of a local public good. With only two workers, (w^*, H^*) can be chosen to keep both workers on their respective indifference curves, but now, w^* will exceed w_0 and the employee is not free to choose hours. Becker (1985, p. s43) noted that “Firms buy a package of time and effort from each employee with the payment tied to the package”. An employee is unable to optimize along each dimension of the package. If there are three or more workers who must conform to a two-dimensional package, (w, H) , at least one worker will almost surely enjoy an economic rent. The magnitude of the rents will be larger, the greater is the dispersion in tastes and the bigger is the team.³⁹ The added

³⁸ The respondent was asked, “Is there a group of people that you think of as your co-workers, people whom you see just about every day and with whom you have to work closely in order to do your job?”. If yes, “About how many people are there in this group?” If there were two teams with 5 and 10 members, the mean size using team weights would be $(5 + 10)/2 = 7.5$, while the mean size using worker weights would be $(5 \times 5 + 1 \times 10)/15 = 8.33$. The mean reported here is a worker-weighted mean.

cost of compensating employees to conform to prescribed work rules is more than offset by the increasing returns to volume and to team production.

4.5. Durable jobs and human capital

A good can be produced in a variety of ways. One could build a steel and concrete warehouse that will last for years or stack the goods in a wooden shack. An investment in an irrigation network is land-specific. Its durability depends on a host of variables including the expected period of production, the transferability of the network to another firm, the cost of "built-in" durability. Bohm-Bawerk believed that roundabout methods of production were more efficient, but that conclusion surely depends on the discount rate. An employer can purchase only the services of labor by turning to the temporary help industry. Most firms, especially larger ones, allocate resources to assemble a work force and to enhance the human capital of their employees in several ways. First, they establish internal labor markets so that employees can have opportunities for career growth within the firm. Second, a higher firm survival rate enhances the returns to training and the amount invested in it. Third, prices of non-labor inputs are lower for larger firms which lead to a higher capital to labor ratio as well as a higher ratio of mobility inhibiting fringe benefits to wages. Mincer (1962) introduced the idea that human capital is produced via on-the-job training. That it is optimal to share the returns was demonstrated by Hashimoto and Yu (1980). The firm and worker are partners in the decision to "produce" human capital. This decision affects not only the amount invested in OJT but also other aspects of the employment relation, the design of the pay package, recruiting practices, job security, and promotions. The literature has mainly embraced the dual labor market paradigm which emphasizes the dichotomy of primary and secondary markets and ignores the continuous variations in firm size.

4.5.1. On-the-job training

The chances of receiving any of five different kinds of training (formal, informal, by co-workers or managers) were higher in larger establishments and in multi-plant firms (Baron et al., 1987). The available international evidence goes in the same direction. In Canada, non-apprenticeship training programs were longer in larger firms, Simpson (1984). Schaffner (1996) reports on data from Peru where the percentage of workers completing OJT climbed with establishment size.

The gap in training investments between large and small employers is larger at higher skill levels; see Haber et al. (1988, Table 8). The training at larger firms is partially

³⁹ The need to compensate employees to conform to the local public goods provided by an employer was developed by Stafford (1980). With two workers and two instruments (a wage rate and weekly hours), a firm can choose (w, H) so that each worker remains on his/her indifference curve. With three members, an employer can recapture part of any rent by introducing a third instrument such as a fringe benefit. Increasing firm size is accompanied by a growing complexity of the compensation package which is consistent with a rent recapture thesis, a point developed by Oi (1990) in footnote 29.

directed at molding workers into teams that can work together and to learn how to operate in a bureaucracy. Haber et al. (1988) found that the ratio of on-site to off-site training is higher in larger firms suggesting that a larger fraction of the human capital investments made by large firms is firm-specific.

4.5.2. *The wage-tenure profile*

On-the-job training is not easily distinguished from learning through experience. Both can contribute to a worker's stock of general or firm-specific capital. If the former, an employer has to raise the employee's wage to retain her. If the investment in firm-specific capital is shared, the employee forgoes earnings during the training period to be repaid later in his tenure with the firm. A third reason for an upward sloping wage-tenure profile is the back loading of pay, the practice of promising a prize to those who stay with the firm (Lazear, 1981). The cross-sectional data generally reveal steeper slopes at larger firms.⁴⁰ The panel data are mixed and often find flatter wage-tenure profiles in larger firms. Baron et al. (1987, p. 87) found that a 10% increase in employer size was associated with a 0.13% reduction in a wage growth ratio measured by $G = (W_2/W_0)$, where W_0 is the starting wage and W_2 is the wage 2 years later. The authors argued that large employers do more screening of job applicants, while small firms evaluate the worker on the job and discharge the poor performers. Investment in entry level training is greater in larger firms which by itself implies a faster rate of wage growth. However, Holtmann and Idson (1995) suggest that the employer share of the training costs is higher in larger firms which operates in the opposing direction. The internal labor market is not a spot market but is supported by implicit contracts. There is no reason to presume that the wage will be proportional to productivity in any short period, but productivity over the contract period must equal or exceed wages plus training costs.

4.5.3. *Durable jobs*

Labor turnover rates are inversely related to firm size and are lower for males. Job tenure in 1993 averaged 4.92 years for men and 2.47 years for women workers. Table 15 reveals that jobs are more durable in larger firms and establishments. Employers who incur larger overhead costs for training and hiring have an obvious incentive to reduce turnover through lower discharge and layoff rates. Workers also incur fixed costs in the search for a job. The higher ratio of applicants to job vacancies means that these search costs are higher for jobs located in larger firms. Quit rates are lower not only to amortize the fixed search costs but also because workers can substitute intra-firm for inter-firm mobility to jobs in different occupations. Wages at big firms contain post-contractual rents, especially for infra-marginal team members. This is confirmed by the responses to the question,

⁴⁰ Pearce (1990) found that in the US non-union wage-tenure profiles are steeper in larger establishments. However, in unionized plants where workers are earning rents, the wage-tenure profiles are flatter. Hashimoto and Raisian (1985) documented steeper wage-tenure profiles in large Japanese firms. For Canada, Reilly (1995) found greater average wage growth in larger establishments in some years, but no size differences in other years.

Table 15
Average tenure by firm and plant size, 1983 and 1993^{a,b}

	SF ^c	SP/LF	LP/LF
1983			
Male	4.98	8.47	11.53
Female	4.19	4.72	7.14
Union	6.85	10.80	11.86
Not union	4.52	6.16	8.66
College	4.78	7.24	10.74
High school	4.73	7.11	8.84
1993			
Male	6.34	9.24	12.06
Female	5.67	6.34	8.87
Union	7.10	14.22	14.33
Not union	5.98	7.42	9.63
College	6.37	8.17	11.66
High school	5.92	8.35	8.95

^a Source: Current Population Survey.

^b Average number of years of current tenure are reported, not the duration of completed spells of employment with the firm (private sector).

^c SF = firm size 1–24, SP/LF = plant size 1–24 and firm size 1000 or more, LP/LF = plant and firm size 1000 or more.

“About how easy would it be for you to find a job with another employer with approximately the same income and fringe benefits that you now have?”. Those at small firms indicated more affirmative answers.⁴¹ The inverse relation between turnover and firm size is more carefully documented by Idson (1996).

The flex-wage model of Pissarides (1977) appears to describe the behavior of a small employer who accepts the first job applicant and offers him a wage equal to his expected marginal value product. It is consistent with a low monitoring cost and a short horizon. Performance is easily observed. Tasks can be adjusted to fit the employee's abilities. If he is unsatisfactory, he can be dismissed, and the process repeated. If by chance, a firm lands a good worker, pay can be raised, and the workload expanded. Small firms have shorter life expectancies. A shorter payout period reduces the returns to investments in specialized equipment or in searching for qualified workers.

A large employer announces a fixed wage for a position, establishes qualification standards, and searches until a suitable candidate is found. Volume production is achieved

⁴¹ The responses were coded from 1 = very easy to 3 = not at all easy. The mean and standard deviation for those in establishments with less than 50 workers were $m = 1.937$ and $s = 0.784$, compared to $m = 2.399$ and $s = 0.716$ at establishments with more than 1000 workers. (Source: 1977 Quality of Employment Survey).

by combining specialized equipment with trained labor. According to Idson (1996), large firms have the capacity to provide durable jobs for two reasons: (1) the ability to substitute intra-firm mobility for inter-firm mobility; and (2) lower firm failure rates. To these, we may add (3) the adoption of a wage policy to encourage low labor turnover.

To sum up, the firms that get and remain “large” organize production around teams, divisions, and profit centers. Capital is substituted for labor. The capital is newer, more durable, and more intensively utilized. A large organization also acquires a higher quality work force. The individuals who are recruited, promoted, and retained possess observable personal characteristics associated with greater productivity. Education, work experience, marital status, willingness to accept a full-time position, are proxies for “built-in” productivity which is approximated by the concept of general human capital. Some firms choose to enhance the human capital of its employees by investing in training. The amount invested in training by the firm and worker (denote this by I) is positively related to firm size. The training investment and the returns to that investment depend in an obvious way on Q = the quality of the trainee (“built-in” productivity), T = the expected period of employment (durability of the employment relation), and r = the discount rate. The back loading of pay provides an incentive to stay with the firm and will be incorporated into the pay package where the training investment I is large. The resources devoted to training will realize a higher return by hiring higher quality trainees and reducing attrition rates. The literature is rich in supplying us with evidence on various pair-wise relations. It is clear, however, that the discount rate r is inversely related to firm size. Given r , training, “built-in” worker quality, and completed job tenure $[I, Q, T]$ are determined by a system of simultaneous equations. The returns to the increment to human capital that was “produced” by the employer-employee attachment will be shared. The wages of workers in larger firms are higher because these workers are simply more productive.

5. Why does employer size matter?

This chapter dealt with the determinants of wages for jobs that are differentiated by employer size. Moore (1911, p. 163) concluded: “Our investigation has yielded the definite result that, as the size of the establishment increases, the condition of the laborer improves in all directions”. Nearly a century later, the facts have not changed. A worker who holds a job in a large firm is paid a higher wage, receives more generous fringe benefits, gets more training, is provided with a cleaner, safer, and generally more pleasant work environment. She has access to newer technologies and superior capital equipment. She is, however, obliged to produce standardized as opposed to customized goods and services, and for the most part to perform the work in tandem with other members of a larger team. The costs of finding a job with a small firm are lower. The personal relation between employee and employer may be closer, but layoff and firm failure rates are higher resulting in less job security.

The size of a firm depends on external market forces, the decisions made by the

entrepreneur, and luck. A firm becomes large and retains its size by organizing production and assembling the right work force. The characteristics of jobs and of the employees who staff them are jointly determined by the employer and employees. The implicit employment contracts result in a reduced form relation in which wages are positively related to size. A monitoring cost hypothesis matches productive workers with large firms to minimize the sum of wage and monitoring costs. Efficiency wages might be paid to deter shirking. Both explanations are vague about why the frequency and virulence of shirking would have been greater in larger firms in the absence of monitoring or efficiency wages. After reviewing a number of theoretical models and the empirical evidence, Brown and Medoff (1989, pp. 1056–1057) concluded:

Our analysis leaves us uncomfortably unable to explain it.... In lieu of a more positive conclusion, we offer two observations... First, large employers pay more for their labor but less for their other inputs because of lower interest rates on borrowed funds and quantity discounts on purchased inputs... Second, large firms are also older firms. Is it possible that the size-wage premium is really a relationship between employer age and wages?... Thus, the employer size-wage effect remains a fact in need of an empirically based theory.

The surplus of revenues over labor costs per employee is positively related to firm size if a part of the income of the owner-manager of a small firm is properly counted as a labor cost. At least three factors could produce this relation of a greater ability to pay by a larger firm: (1) lower prices for non-labor inputs; (2) greater market power; or (3) larger overhead cost to amortize sunk costs for capital and a firm-specific work force. It is important to identify the source of the surplus. If it is a true economic rent (due to monopoly power, luck, or a fortuitous location), workers could capture a portion by forming a trade union. Rent sharing may be practiced by an altruistic firm, but our received theory is silent about which firms will be altruistic. There are other instances in which a post-contractual rent represents a return to the firm for an investment in human capital. The gross returns to firms and employees could be increased by sharing the returns. A specifically trained employee thus earns a wage that exceeds her wage in an alternative employment and hence has a lower propensity to separate. One could either call this rent-sharing or label it as a means of “producing” a more productive labor input. The relation of wages to “firm age” is weaker. The merits of this argument are examined in Appendix A.

The taxation and regulation of business enterprises surely affect the size distribution of firms and hence the shape of the wage–size profile. One can ask whether an industrial policy that favors some sectors over others or a policy of subsidizing small business will be beneficial to labor. The answer will depend on the theory of wages which one accepts. If wages are the results of discretionary wage policies, intervention might well improve labor’s position. This chapter advanced a productivity hypothesis which begins with the premise that labor productivity is a function of work effort. Production, especially in the distributive trades, is characterized by increasing returns. A higher arrival rate of customers, clients, and patients means that workers have less idle time and hence are more

productive at larger establishments. In the goods producing sector, larger firms organize work around teams, establish higher effort standards, recruit, train, and retain more productive employees. Schultze (1989, p. 280) asks us to think about two 30-year-old college graduates, one manages a Kentucky Fried Chicken outlet, and the other works at the World Headquarters of the Ford Motor Company. The two are not interchangeable. Productivity is a function of both observable traits as well as unobservable characteristics. Jobs at large firms are different from those at small firms. The organization of work and the selection of employees are responsible for the fact that wages are positively related to employer size. This view of the labor market argues against interventions to manipulate the size distribution of firms.

Appendix A. On the relationship of wages to firm age

In the conclusion to their survey, Brown and Medoff (1997) noted that large firms are older which led them to ask, “Is the size–wage premium really a relationship between employer age and wages?”. The 1987 Survey of Manufactures asked for the earliest date at which production commenced at this establishment. Dunne and Roberts (1990) could thus introduce plant age as a determinant of wages (average hourly earnings of production workers). In a wage equation which included establishment size, capital stock, industry, and region, they found that plants, 30 or more years old, paid wages which were 10% higher than wages at plants 1 to 5 years old.⁴² This result might have been caused by the possibility that older plants had more experienced workers. Troske (1994) found that workers in plants less than five years old earned 20% less than workers in plants in business for fifteen or more years. However, controlling for worker characteristics in his sample of manufacturing workers reduced the partial effect of plant age on wages. A more careful analysis of this relationship was undertaken by Brown and Medoff (1997). They attached additional questions to the monthly Survey of Consumers to obtain data on age of firm, probability of layoff or plant closure, and the employer’s ability to pay.⁴³ In a linear model, a one standard deviation increase in firm age was associated with a 7% increase in wages.⁴⁴ A plot of the data revealed that the relationship was U-shaped. Wages declined with firm age, but for firms above the median “age” of 30 years, wages were positively related to firm age. Controlling for worker/job characteristics extended the negatively sloped segment. The authors advance several reasons to rationalize the downward sloping segment of the

⁴² Establishments that were part of a multi-plant firm paid wages 21% above wages of single plant firms. There is some evidence that wages were negatively related to the exit and entry rate of plants into the industry.

⁴³ This last was calculated from responses to the question, “Without going out of business, do you think that your employer could afford to increase the pay of all its workers by 10%?”. The questions on firm age and layoff probability are reproduced in note 6 of Brown and Medoff (1997) who collected data on worker and job traits including firm and plant size. Of the 1410 workers in private and non-profit sectors from September 1991 to March 1992, data on firm age were provided by 1076 respondents.

⁴⁴ The mean of the “firm age” variable in their sample was 40.1 years with a standard deviation of 32.0 years. In logs, the mean was 3.29 with a standard deviation of 1.09.

age-wage profile. We shall remark on two of these. First, older firms are more likely to offer fringe benefits. We are a bit confused by the chain of reasoning in the Brown-Medoff paper. "One reason is that as noted above, they tend to pay higher wages, and such fringes have a tax advantage...". Newly established firms are uncertain of their survival and choose to postpone adopting fringe benefit plans which have a significant fixed set-up cost. An older established firm can then substitute higher fringe benefits for lower wages. The relationship to be explained is that between wages and firm age, not total hourly compensation including fringes and firm age. Second the evidence is overwhelming that older firms have higher survival rates. The Survey of Consumers respondents evidently indicated that the odds of layoffs or plant closures were lower at older firms. The authors assume that workers will accept a lower wage in return for greater job security. This assumption was challenged by Mayo and Murray (1991) and Winter-Ebmer (1995) whose studies indicated that increases in the probability of job loss were accompanied by lower wages. Finally, we question the authors who wrote, "...there is undoubtedly a positive relationship between the age of the business and the age of the capital it uses. It is possible that older firms use older capital which requires less skill and so employ less skilled employees". Dom et al. (1993) found that the adoption of advanced technologies was more likely among large and hence older manufacturing plants which also hired more skilled workers and paid them higher wages. The giant supermarkets occupied newer buildings and operated more sophisticated equipment.

The upward sloping portion of the age-wage relationship was rationalized by the greater ability to pay by very old firms over 30 years of age. "Ability" in this study is based on the workers' assessment of whether the employer could raise pay by 10% without going out of business. If one could get data on profits per worker, which is an alternative and possibly better proxy for "ability to pay", its inclusion in a wage equation should do away with the positive branch of the U-shaped profile.

The firm is at best a fuzzy entity in our received theory of price, especially when one has to cope with joint ventures, multi-product "firms", partnerships, and so on. A recurring theme in Marshall (1952) is the non-constancy of the human agent. The situation is even more fluid for a firm in a changing world. The company that produced railway signals in 1920 is still in business in Rochester, but it no longer makes railway signals. Is it still the same *firm*? If Yamaha closes all of its motorcycle factories and deals only in the production and distribution of pianos, do we measure its age from the start of the motorcycle or piano eras? Acquisitions and divestures pose a hard problem. San Michelle Wines was acquired by US Tobacco. Does an employee at the vineyard report the age at which San Michelle began bottling wine or the age at which US tobacco began to make pipe tobacco? "When a business is sold to a new owner, Dunn and Bradstreet's "years in business" variable often measures years in business under current ownership" (cited in Brown and Medoff, 1997). The cross-sectional samples examined by Dunne and Roberts (1990) and by Brown and Medoff (1997) yield conflicting relationships. A strong test of the relationship of wages to firm age would require panel data in which one could follow wages as a

firm gets older. We do not have such data, and given the elusive nature of the concept of a firm, it is unclear if we want to wait for someone to collect such a panel dataset.

References

- Akerlof, G.A. (1984), "Gift exchange and efficiency-wage theory: four views", *American Economic Review Papers and Proceedings* 74: 79–83.
- Alchian, A.A. (1959), "Costs and outputs" in: M. Abramovitz, ed., *The allocation of economic resources* (Stanford University Press, Stanford, CA).
- Alchian, A.A. and H. Demsetz (1972), "Production, information cost, and economic organization", *American Economic Review* 62: 777–795.
- Asher, H. (1956), "Cost quantity relationship in the air frame industry", Report no. R-291 (RAND, Santa Monica, CA).
- Autor, D.H., L.F. Katz and A.B. Krueger (1997), "Computing inequality: have computers changed the labor market?" Working paper no. 5956 (NBER, Cambridge, MA).
- Baker, G.P., M. Jensen and K.J. Murphy (1988), "Compensation and incentives: practice vs. theory", *Journal of Finance* 43: 593–615.
- Baldamus, W. (1959), *Efficiency and effort* (Tavistock Publications, London).
- Baron, J., D.A. Black and M.A. Loewenstein (1987), "Employer size: the implications for search, training, capital investment, starting wages, and wage growth", *Journal of Labor Economics* 5: 76–89.
- Becker, G.S. (1973), "A theory of marriage: part I", *Journal of Political Economy* 81: 813–846.
- Becker, G.S. (1985), "Human capital, effort, and the sexual division of labor", *Journal of Labor Economics* 3 (part 2): S33–S58.
- Beecher, J. (1986), *Charles Fourier, a visionary and his world* (University of California Press, Berkeley, CA).
- Bluestone, B. (1989), *The retail revolution* (The Auburn House, Boston, MA).
- Brown, C. and J. Medoff (1989), "The employer size wage effect", *Journal of Political Economy* October: 1027–1059.
- Brown, C. and J. Medoff (1997), "Firm age and wages", Working paper.
- Brown, C., J. Hamilton and J. Medoff (1990), *Employers large and small* (Harvard University Press, Cambridge, MA).
- Bulow, J.I. and L.H. Summers (1986), "A theory of dual labor markets with applications to industrial policy", *Journal of Labor Economics* 4: 376–414.
- Chandler, Alfred D. Jr. (1982), *The visible hand* (Harvard University Press, Cambridge, MA).
- Clark, J.M. (1923), *Studies in the economics of overhead costs* (University of Chicago Press, Chicago, IL).
- Davis, S. and M. Henrekson (1997), "Explaining national differences in the size and industry distribution of employment", Working paper (August).
- de Grazia, Sebastian (1962), *Of time, work, and leisure* (Twentieth Century Fund, New York).
- Deardorf, A.V. and F.P. Stafford (1976), "Compensation of cooperating factors", *Econometrica* 44: 671–684.
- Doeringer, P. and M. Piore (1971), *Internal labor markets and manpower analysis* (D.C. Heath, Lexington, MA).
- Doms, M., T. Dunne and K.R. Troske (1997), "Workers, wages, and technology", *Quarterly Journal of Economics* 102: 253–290.
- Dunn, L.F. (1980), "The effect of firm and plant size on employee well-being", in: J.J. Siegfried, ed., *The economics of firm size, market structure, and social performance* (US Government Printing Office, Washington, DC) pp. 348–358.
- Dunne, T. and M. Roberts (1990), "Plant, firm and industrial wage variations", Working paper (Pennsylvania State University, December 1990).
- Dunne, T. and J. Schmitz Jr. (1995), "Wages, employment structure, and employer size wage premia: their relationship to advanced technology usage at U.S. manufacturing establishments", *Economica* 62: 89–105.

- Eaton, B.C. and W.D. White (1983), "The economy of high wages: an agency problem", *Economica* 50: 175–181.
- Ehrenberg, R.G. and R.S. Smith (1997), *Modern labor economics* (Addison-Wesley Educational Publishers, Redding, MA).
- Evans, D. and L. Leighton (1989), "Why do smaller firms pay less?" *Journal of Human Resources* 24: 299–318.
- Farber, H.S. (1997), "Job creation in the United States: good jobs or bad?" Working paper no. 385 (Princeton University, Princeton, NJ).
- Foss, M.R. (1981), *Changes in the workweek of fixed capital* (American Enterprise Institute, Washington, DC).
- Funkhauser, E. (1997), "The importance of firm wage differentials in explaining hourly earnings variations in a large scale sector of Guatemala", *Journal of Development Economics* 513: 1–17.
- Garen, J. (1985), "Worker heterogeneity, job screening, and firm size", *Journal of Political Economy* 93: 715–739.
- Gibbons, R., and L. Katz (1992), "Does unmeasured ability explain inter-industry wage differentials?" *Review of Economic Studies* 59: 515–535.
- Griliches, Z. (1969), "Capital-skill complementarity", *Review of Economics and Statistics* 51: 465–468.
- Groschen, E.L. (1991), "Five reasons why wages vary among employers", *Industrial Relations* 30: 350–381.
- Groschen, E.L. and A.B. Krueger (1990), "The structure of supervision and pay in hospitals", *Industrial and Labor Relations Review* 43: S134–S146.
- Haber, S., J. Cordes and J. Barth (1988), "Employment and training opportunities in small and large firms", Report (U.S. Small Business Administration, Office of Advocacy).
- Hall, R.E. (1982), "The importance of lifetime jobs in the U.S. economy", *American Economic Review* 72: 716–724.
- Hall, R.E. (1991), *Booms and recessions in a noisy economy* (Yale University Press, New Haven, CT).
- Hamermesh, D. (1993), *Labor demand* (Princeton University Press, Princeton, NJ).
- Hashimoto, M. and J. Raisian (1985), "Employer tenure and earnings profiles in Japan and the United States", *American Economic Review* 75: 721–735.
- Hashimoto, M. and B.T. Yu (1980), "Specific capital, employment contracts, and wage rigidity", *The Bell Journal of Economics* 11: 536–549.
- Höftmann, A.G. and T.L. Idson (1995), "Information, employer size, training, and wage growth", *Eastern Economics Journal* 21: 187–191.
- Hutt, W.H. (1977), *The theory of idle resources* (Liberty Press, Indianapolis, MN).
- Idson, T.L. (1995), "Team production effects on earnings", *Economic Letters* 49: 197–203.
- Idson, T.L. (1996), "Employer size and labor turnover", in: S. Polachek, ed., *Research in labor economics* (JAI Press, Greenwich, CT).
- Idson, T.L. and D. Feaster (1990), "A selectivity model with employer size differentials", *Journal of Labor Economics* 8 (part 1): 99–122.
- Idson, Todd L. and Hisako Ishii (1992), "Comparison of employer size effects on wages and tenure for men and women in Japan and the United States", *Industrial Relations Research Association Proceedings*: 531–538.
- Kaldor, N. (1934), "The equilibrium of the firm", *Economic Journal* 44: 60–76.
- Katz, L.F. and L.H. Summers (1989), "Industry rents: evidence and implications", *Brookings Papers on Economic Activity, Microeconomics*: 209–275.
- Kent, A.S.S. (1915), *Second interim report on industrial fatigue* (London).
- Kostick, P. (1990), "Firm size and executive compensation", *Journal of Human Resources* 25: 90–105.
- Kremer, M. (1993), "The O ring theory of economic development", *Quarterly Journal of Economics* 108: 551–575.
- Krueger, A. (1993), "How computers changed the wage structure", *Quarterly Journal of Economics* 108: 33–60.
- Kruse, D. (1992), "Supervision, working conditions, and the employer size effect", *Industrial Relations* 31: 229–249.
- Kwoka, J.E. Jr. (1980), "Establishment size, wages, and job satisfaction, the trade-offs", in: J.J. Siegfried, ed., *The economics of firm size, market structure, and social performance* (US Government Printing Office, Washington, DC) pp. 359–379.

- Lazear, E. (1981), "Agency, earnings profiles, productivity, and hours restrictions", *American Economic Review* 71: 606–620.
- Lester, R.A. (1967), "Pay differentials by size of establishment", *Industrial and Labor Relations Review* 7: 57–67.
- Little, I.M.D., D. Mazundar and J.M. Page Jr. (1987), *Small manufacturing enterprises, a comparative analysis of India and other economies* (Oxford University Press, Oxford, UK).
- Loveman, G. and W. Sengenberger (1991), "The re-emergence of small scale production: an international comparison", *Small Business Economics* 3: 1–37.
- Lucas, R.E. Jr. (1978), "On the size distribution of business firms", *The Bell Journal of Economics* 9: 508–523.
- Main, B.G.M. and B. Reilly (1993), "The employer size-wage gap: evidence for Britain", *Economica* 60: 125–142.
- Marshall, Alfred (1952), *Principles of economics* (Macmillan, New York).
- Mayo, J.W. and M.N. Murray (1991), "Firm size, employment risk, and wages: further insights on a persistent puzzle" *Applied Economics* 23: 1351–1360.
- McNulty, P.J. (1980), *The origins and development of labor economics* (M.I.T. Press, Cambridge, MA).
- Mellow, W. (1982), "Employer size and wages", *Review of Economics and Statistics* 54: 495–501.
- Mincer, J. (1962), "On-the-job training: costs, returns, and some implications", *Journal of Political Economy* 70: s50–s79.
- Moore, Henry L. (1911), *Laws of wages* (Macmillan, New York) (reprinted 1967, Augustus M. Kelley, New York).
- Mulligan, J.G. (1983), "The economies of massed reserves", *American Economic Review* 15: 725–734.
- Nyland, C. (1989), *Reduced worktime and the management of production* (Cambridge University Press, New York).
- Oi, W.Y. (1967), "The neoclassical foundations of production functions" *Economic Journal* 77: 579–594.
- Oi, W.Y. (1974), "On the economics of industrial safety", *Law and Contemporary Problems* 38: 669–709.
- Oi, W.Y. (1983), "Heterogeneous firms and the organization of production", *Economic Inquiry* 21: 147–171.
- Oi, W.Y. (1990), "Employment relations in dual labor markets", *Journal of Labor Economics* 8 (part 2): s124–s149.
- Oi, W.Y. (1991), "Low wages and small firms", in: R.G. Ehrenberg, ed., *Research in labor economics*, Vol. 12 (JAI Press, Greenwich, CT).
- Oi, W.Y. (1992), "Productivity in the distributive trades", in: Z. Griliches, ed., *Output measurement in the service sectors* (University of Chicago Press, Chicago, IL).
- Pearce, J.E. (1990), "Tenure, unions, and the relationship between employer size and wages", *Journal of Labor Economics* 8: 251–269.
- Personick, M.E. and C.B. Barsky (1982), "White-collar pay levels linked to corporate work force size", *Monthly Labor Review* 105: 23–28.
- Pissarides, C. (1977), *Labor market adjustments* (Cambridge University Press, Cambridge, MA).
- Raff, D.M.G. and L.H. Summers (1987), "Did Henry Ford pay efficiency wages?" *Journal of Labor Economics* 5: S57–S86.
- Reilly, K.T. (1995), "Human capital and information: the employer size-wage effect", *Journal of Human Resources* 30: 1–18.
- Robinson, E.A.G. (1958), *The structure of competitive industry* (University of Chicago Press, Chicago, IL).
- Schaffner, J.A. (1996), "Premiums to employment in larger establishments: evidence from Peru", *Working paper* (Stanford University, Stanford, CA).
- Schultze, C.L. (1989), "Industry rents: comment", *Brookings Papers on Economic Activity Microeconomics*: 280–283.
- Senior, N. (1858), *Political economy*, 4th edition (Augustus M. Kelley, Publisher).
- Shapiro, C. and J.E. Stiglitz (1984), "Equilibrium unemployment as a worker discipline device", *American Economic Review* 74: 433–444.
- Shinohara, M. (1962), *Growth and cycles in the Japanese economy* (Kinokuniya Bookstore Co., Tokyo).
- Simon, H.A. (1947), *Administrative behavior* (Macmillan, New York).

- Simpson, W. (1984), "An Econometric analysis of industrial training in Canada", *Journal of Human Resources* Fall: 435–451.
- Slichter, S.H. (1950), "Notes on the structure of wages" *Review of Economics and Statistics* 32: 80–91.
- Smith, A. (1997), *The wealth of nations*, Cannan edition, Vol. I.
- Stafford, F.P. (1980), "Firm size, workplace public goods, and worker welfare", in: J.J. Siegfried, ed., *The economics of firm size, market structure, and social performance* (US Government Printing Office, Washington, DC) pp. 326–347.
- Stigler, G.J. (1962), "Information in the labor market", *Journal of Political Economy* 70 (part 2): S94–S105.
- Stigler, G.J. and R.A. Sherwin (1985), "The extent of the market", *Journal of Law and Economics* 28: 555–585.
- Troske, K.R. (1994), *Evidence on the employer size wage premium from worker establishment matched data* (Center for Economic Studies, US Bureau of the Census, Washington, DC).
- Troske, K.R. (1999), "The worker-establishment characteristics database", in: J. Haltiwanger and M. Manser, eds., *Labor statistics measurement issues* (University of Chicago Press, Chicago, IL) in press.
- Velenchik, A.D. (1996), "Market power, firm performance and real wage growth in Zimbabwean manufacturing", *World Development* 25: 749–762.
- Vernon, H.M. (1921), *Industrial fatigue and efficiency* (G. Routledge, London).
- Weiss, L. (1966), "Concentration and labor earnings", *American Economic Review* 56: 96–117.
- Winter-Ebmer, R. (1995), "Does layoff risk explain the firm-size wage differential?" *Applied Economics Letters* 2: 211–214.
- Wright, T.P. (1936), "Factors affecting the cost of air frames", *Journal of the Aeronautical Sciences* 3: 122–128.